



CMPT 413/713: Natural Language Processing

Post-training: Reasoning

Spring 2026
2026-03-02

Slides adapted from Anoop Sarkar

Training stages

Piles of
unlabeled
text!



Self-supervised training
LM objective

$$p(w_t | w_{<t})$$

Pre-training

LM training on large, large
amount of data

Pre-training can be
broken into stages (mid-
training)

Text with
“instructions”
and
“responses”

List three fruit
Apple, orange,
banana

Supervised training
LM objective

$$p(w_t | w_{<t}; \text{prompt})$$

Instruction-tuning

Supervised fine-tuning for
instructions

Human
preference data
Data about what
people prefer



Reinforcement
learning

Preference optimization

Align to human
preferences

Post-training

(Can have more iterations)

Training for reasoning

Reasoning models

- Think before responding with the answer
- Reasoning problems requires decomposing the problem into multiple steps (think step-by-step)
- Typically math, coding problems

The screenshot shows a chat interface. At the top right, a blue circle icon is labeled "User". Below it, a grey bubble contains the user's query: "What is the pH of a 0.10 M solution of NH_4F ? The K_a of NH_4^+ is 5.6×10^{-10} and the K_a of HF is 6.8×10^{-4} ." Below the query, a response bubble from the model is shown. It starts with a circular icon containing a spiral, followed by the text "Thought for 4 seconds ▾". Below this, the model's reasoning is shown: "To determine the pH of a 0.10 M solution of ammonium fluoride (NH_4F), we'll need to consider the acid-base properties of its constituent ions when dissolved in water." This is followed by "Step 1: Dissociation of NH_4F " and the chemical equation $\text{NH}_4\text{F} \rightarrow \text{NH}_4^+ + \text{F}^-$. A red arrow points from the text "Full long CoT is hidden from the User" to the "Thought for 4 seconds ▾" text. A blue bracket on the right side of the response bubble points to the text "A summary of the long CoT is provided instead".

User

What is the pH of a 0.10 M solution of NH_4F ? The K_a of NH_4^+ is 5.6×10^{-10} and the K_a of HF is 6.8×10^{-4} .

Thought for 4 seconds ▾

To determine the pH of a 0.10 M solution of ammonium fluoride (NH_4F), we'll need to consider the acid-base properties of its constituent ions when dissolved in water.

Step 1: Dissociation of NH_4F

$\text{NH}_4\text{F} \rightarrow \text{NH}_4^+ + \text{F}^-$

Full long CoT is hidden from the User

A summary of the long CoT is provided instead

Reasoning problems

- Problem that requires step-by-step

Model Input

Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?

Model Output

Natalia sold $48/2 = \langle\langle 48/2=24 \rangle\rangle$ 24 clips in May. Natalia sold $48+24 = \langle\langle 48+24=72 \rangle\rangle$ 72 clips altogether in April and May.

Final Answer: 72

Model Input

Weng earns \$12 an hour for babysitting. Yesterday, she just did 50 minutes of babysitting. How much did she earn?

Model Output

Weng earns $12/60 = \langle\langle 12/60=0.2 \rangle\rangle$ 0.2 per minute. Working 50 minutes, she earned $0.2 \times 50 = \langle\langle 0.2*50=10 \rangle\rangle$ 10.

Final Answer: 10

Model Input

James writes a 3-page letter to 2 different friends twice a week. How many pages does he write a year?

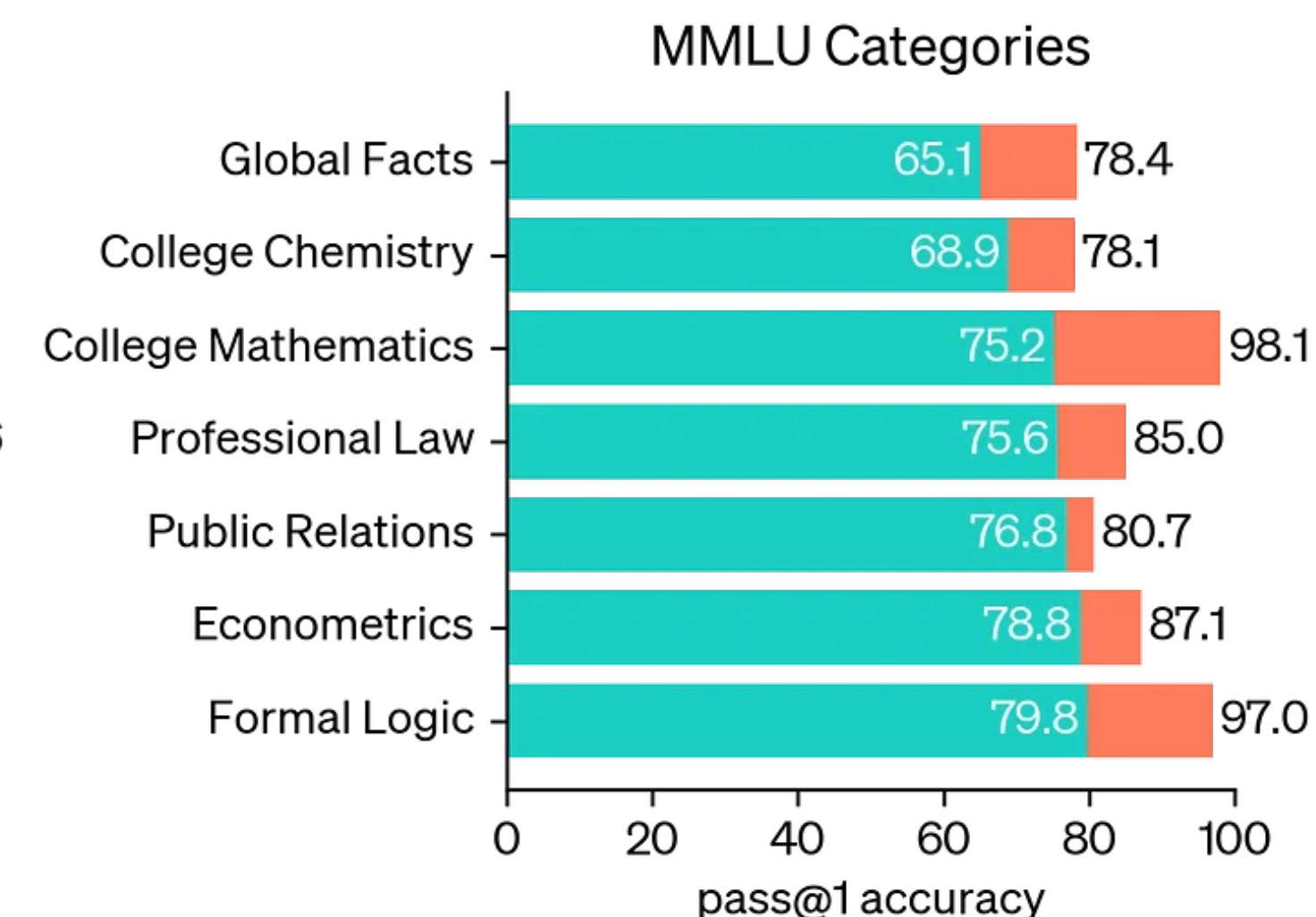
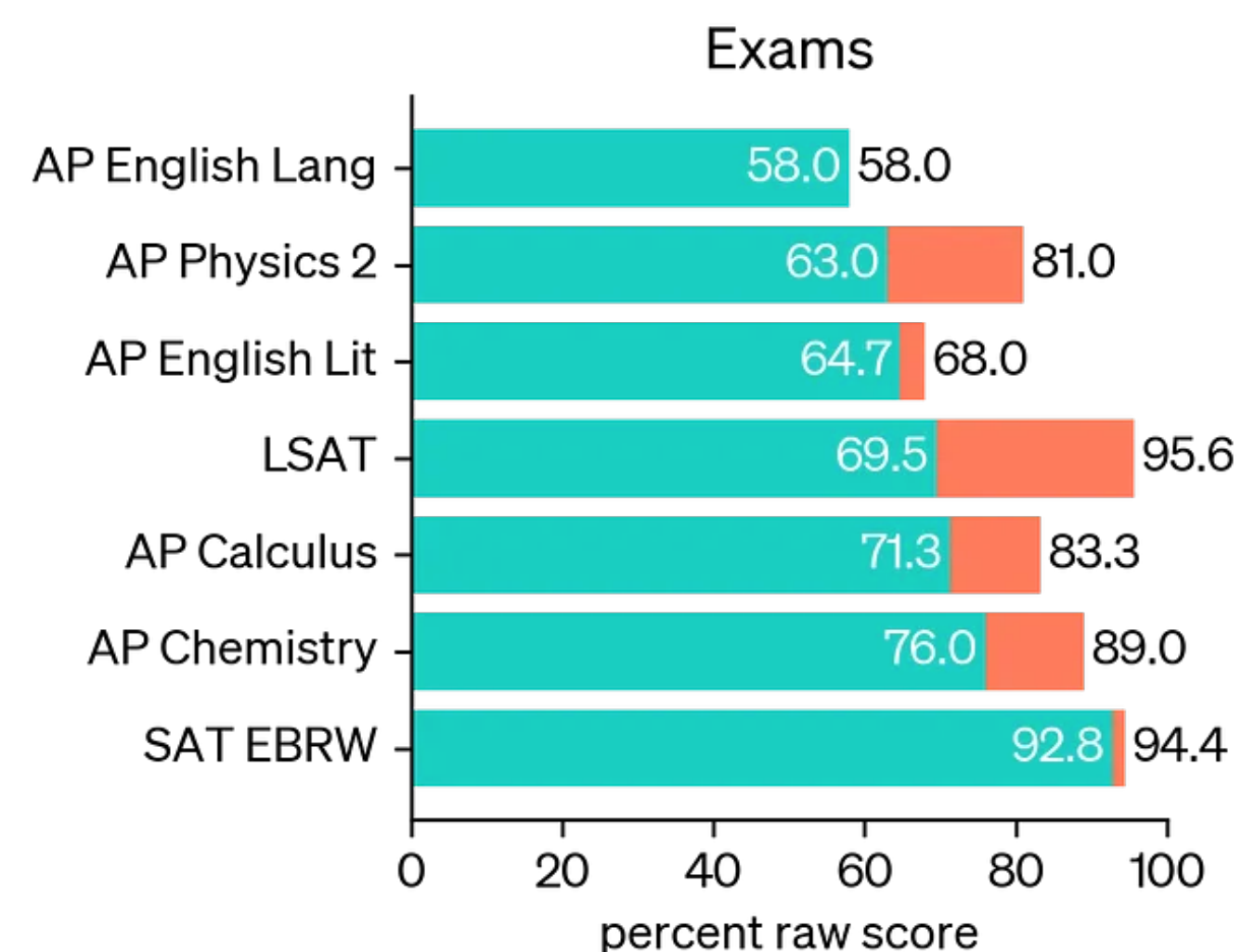
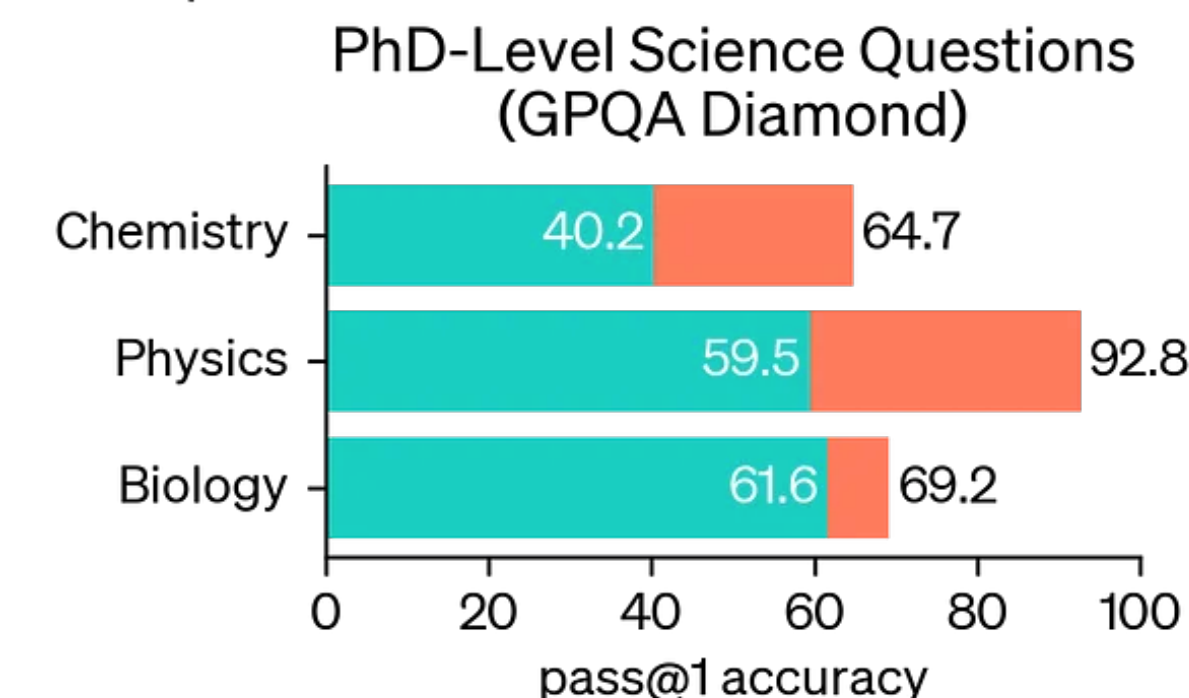
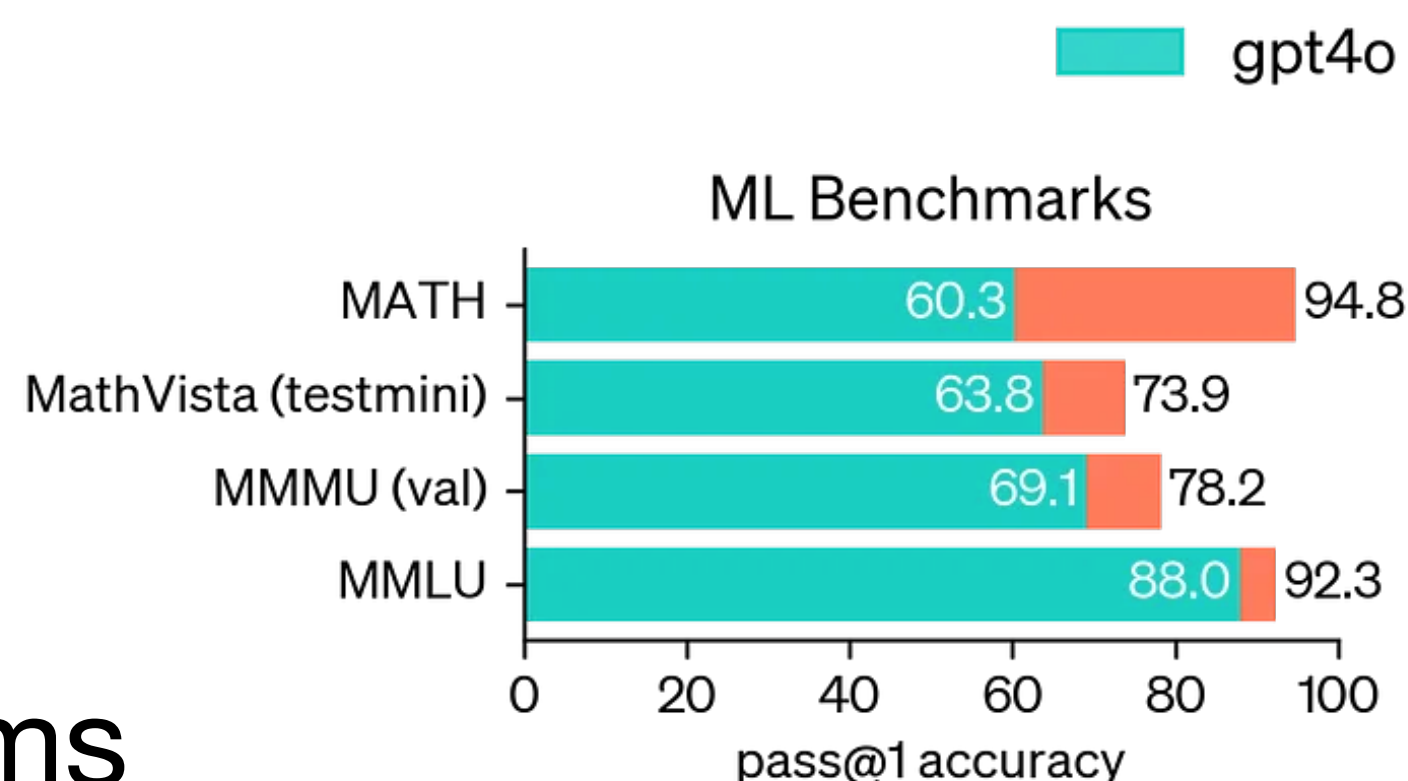
Model Output

He writes each friend $3*2 = \langle\langle 3*2=6 \rangle\rangle$ 6 pages a week. He writes $6*2 = \langle\langle 6*2=12 \rangle\rangle$ 12 pages per week and $12*52 = \langle\langle 12*52=624 \rangle\rangle$ 624 pages a year

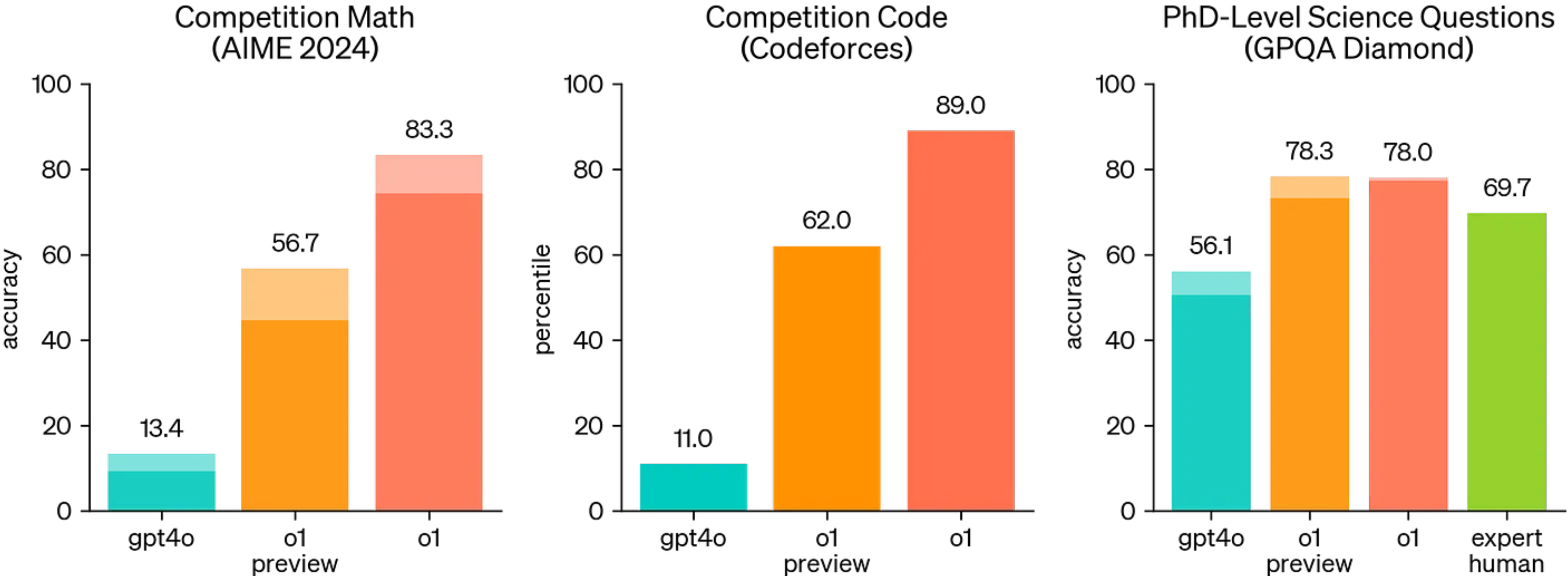
Final Answer: 624

Reasoning models

- Think through each part of a complex problem
- Decompose complex problems into smaller, solvable parts
- Critique its own (partial) solutions and find errors
- Explore many alternative solutions



Reasoning models



Training LLMs to think step-by-step

- Supervised training with reasoning data and reasoning traces
- Reinforcement learning with verifiable rewards (RLVR)

```
<|im_start|>user
Give me a short introduction to large language model.<|im_end|>
```

Prompt / Instruction

```
<|im_start|>assistant
```

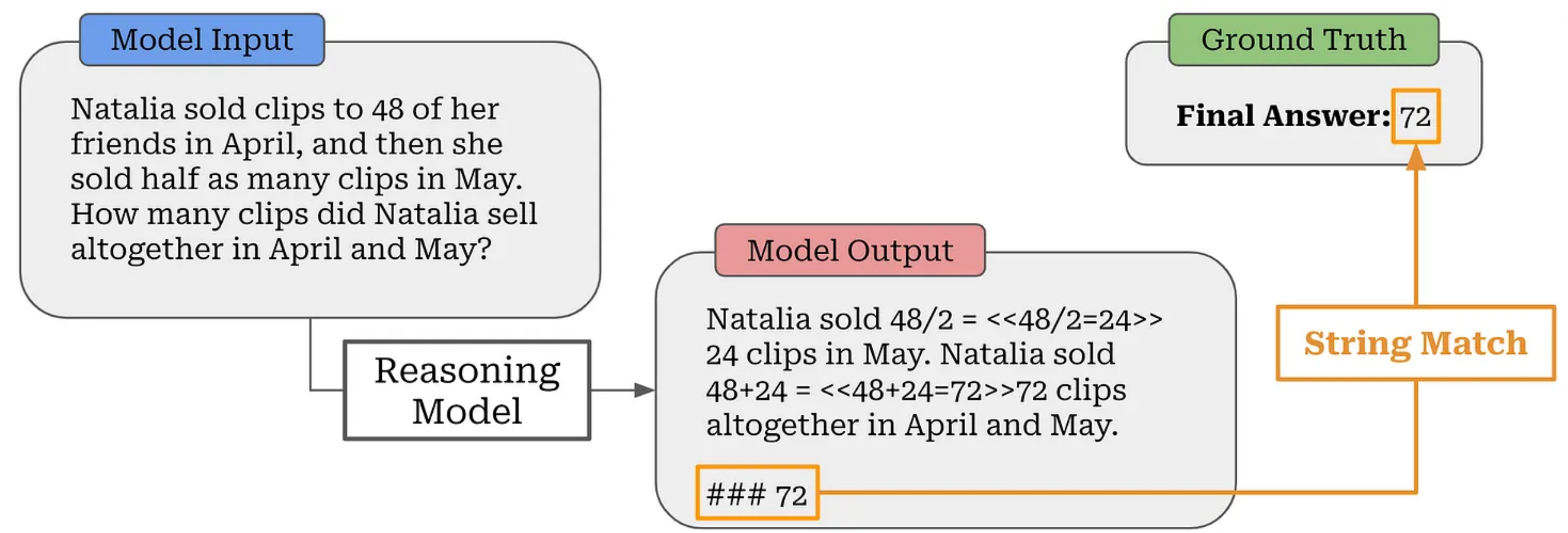
Reasoning Model Output

```
<think>
Okay, the user wants a short introduction to large language
models. Let me start by defining what they are. They're AI
systems trained on massive text data, right? I should mention
their ability to understand and generate human-like text. Maybe
include examples like GPT or BERT. Also, highlight their
applications in tasks like answering questions, writing, and
coding. Keep it concise but cover the key points. Make sure to
note that they're based on deep learning and neural networks.
Avoid technical jargon to keep it accessible. Let me check if
I'm missing anything important. Oh, maybe mention their scale in
terms of parameters. Yeah, that's a common point. Alright, time
to put it all together in a clear, brief paragraph.
</think>
```

Reasoning Trace

```
Large language models (LLMs) are advanced artificial
intelligence systems trained on vast amounts of text data to
understand and generate human-like language. They use deep
learning techniques, such as transformer architectures, to
process and analyze text at scale, enabling them to perform
tasks like answering questions, writing stories, coding, and
translating languages. LLMs are characterized by their massive
parameter counts, which allow them to capture complex patterns
in language, making them highly versatile for a wide range of
applications in natural language processing and beyond.
<|im_end|>
```

Final Response



Training LLMs to think step-by-step

- Supervised training with reasoning data and reasoning traces
- Reinforcement learning with verifiable rewards (RLVR)
- Alternative to training LLMs for reasoning is to scale up test-time compute
 - Reasoning during inference

Olmo training phases

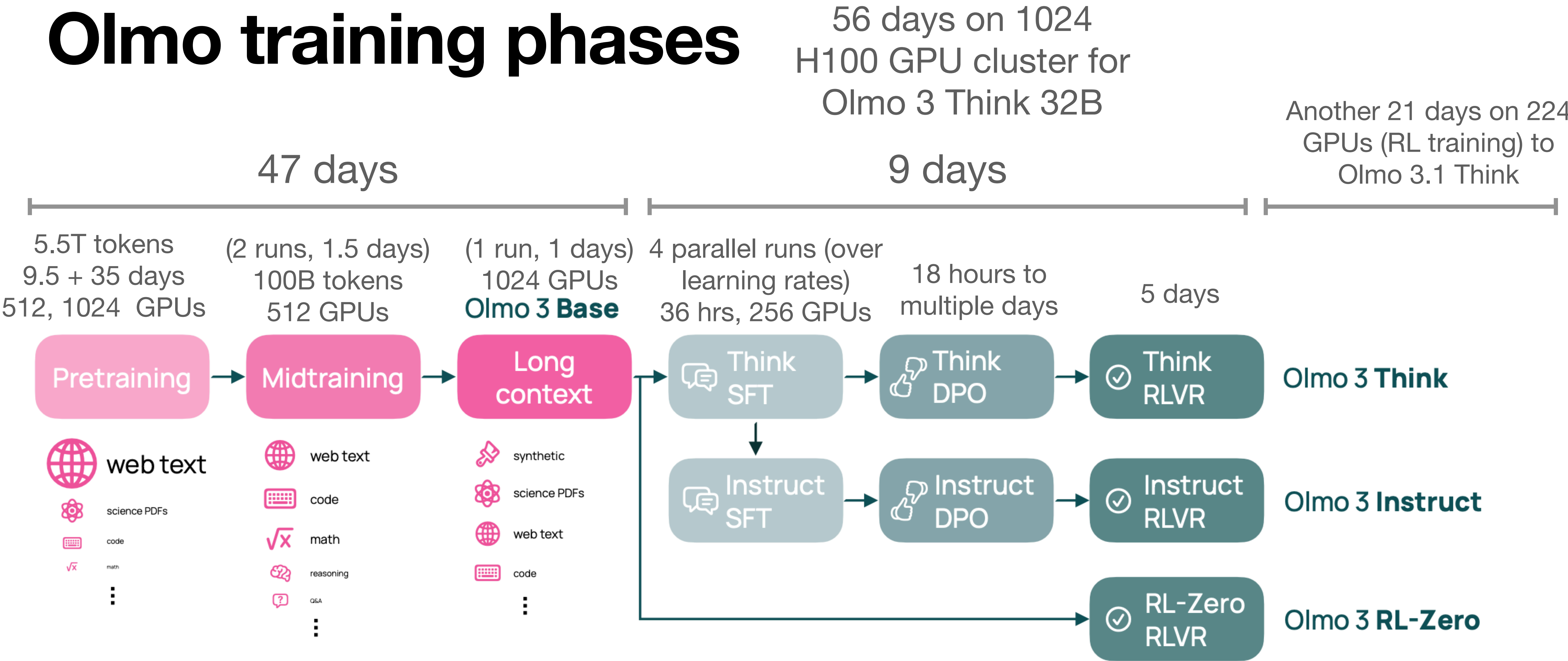
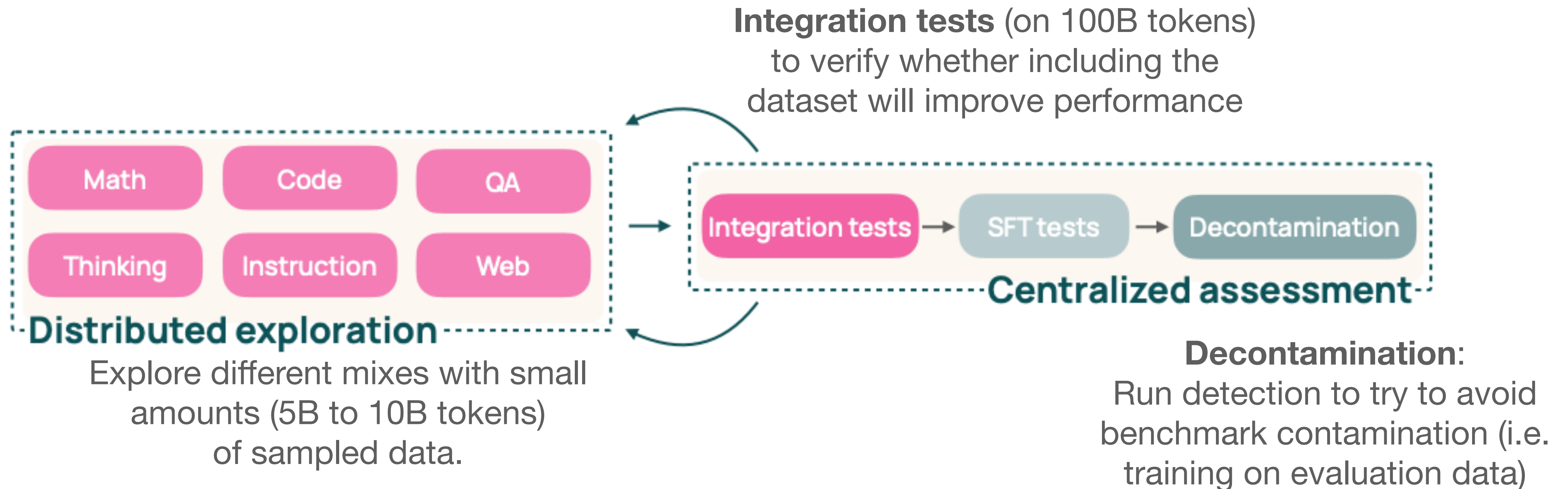


Figure 2 Depiction of model flow for Olmo 3. Development is divided into major **base model training (left)** and **post-training (right)** stages, each further divided into sub-stages with their own recipes (i.e., training data and method).

Figure from <https://arxiv.org/pdf/2512.13961> [Olmo 3, 2025]

Mid-training

Training on cleaned, specialized data



Olmo Post-training

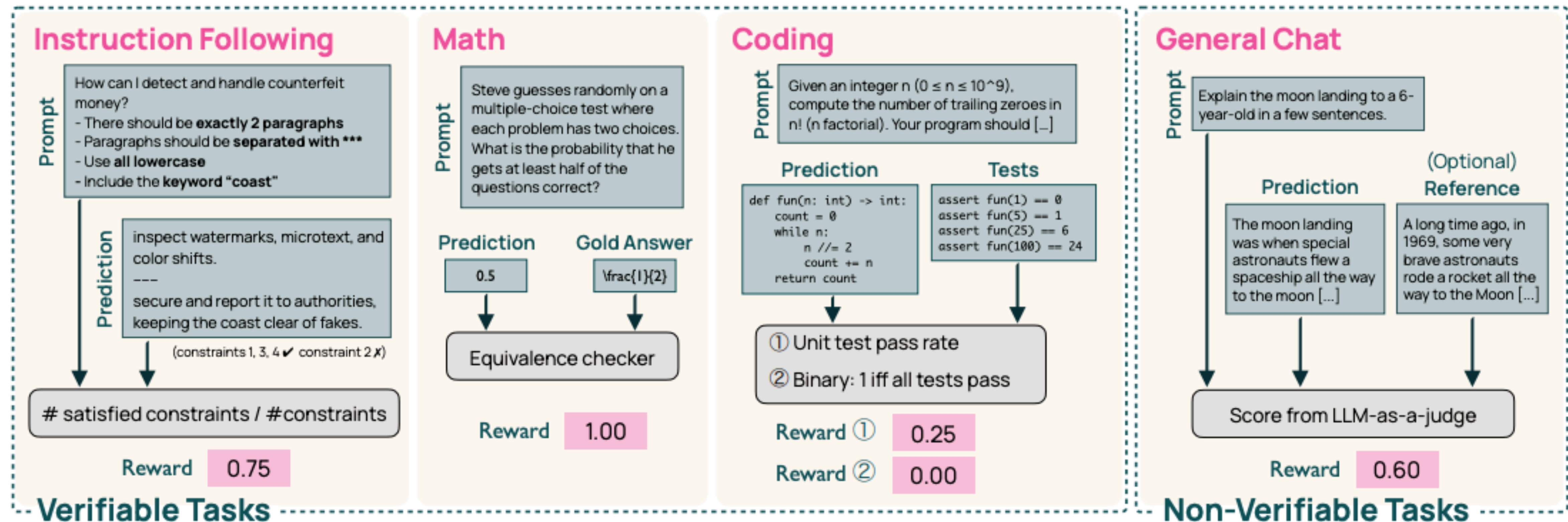
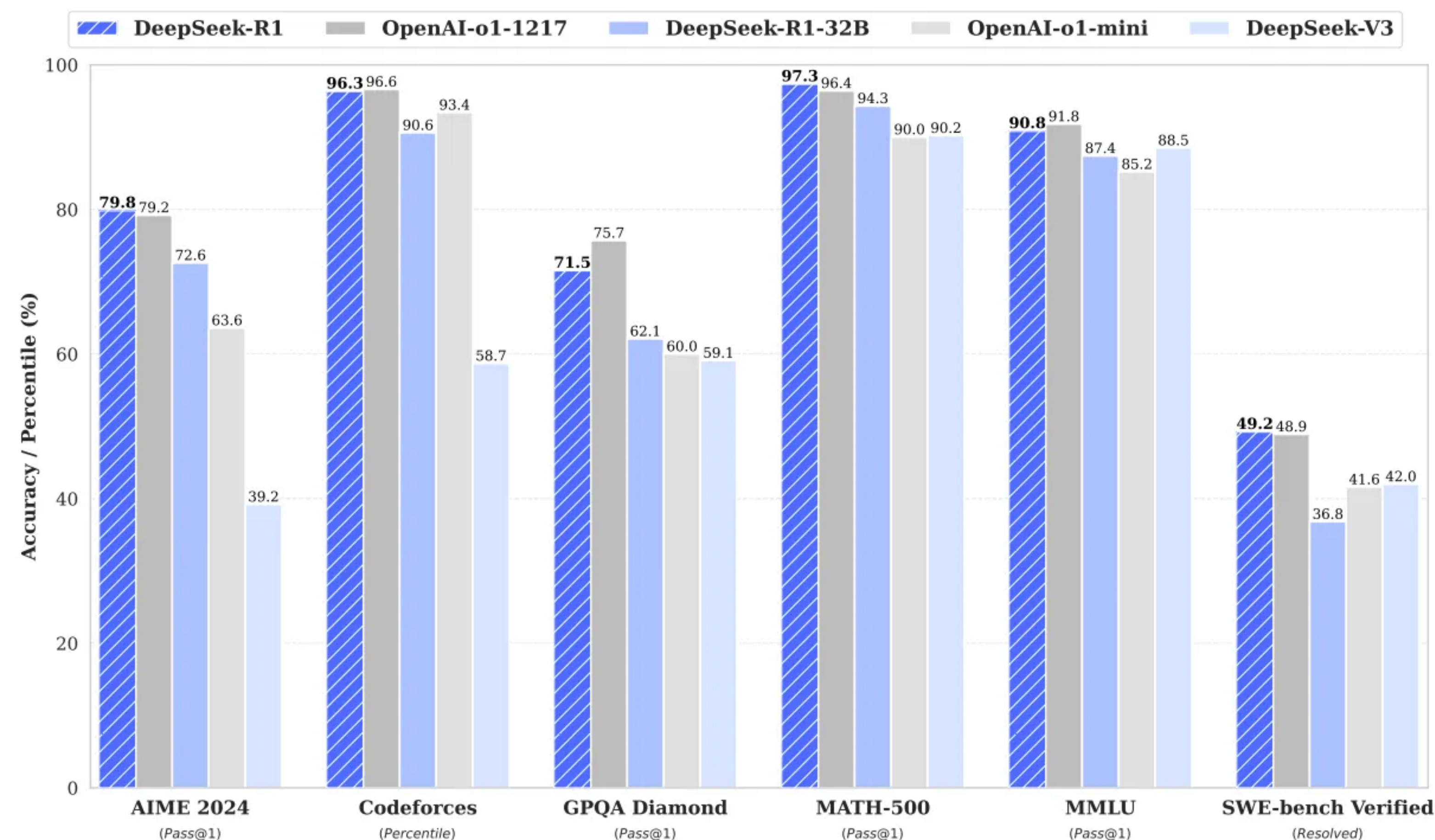


Figure 16 Verifiers and reward design for verifiable and non-verifiable tasks.

Deepseek R1

Competitive open-source model

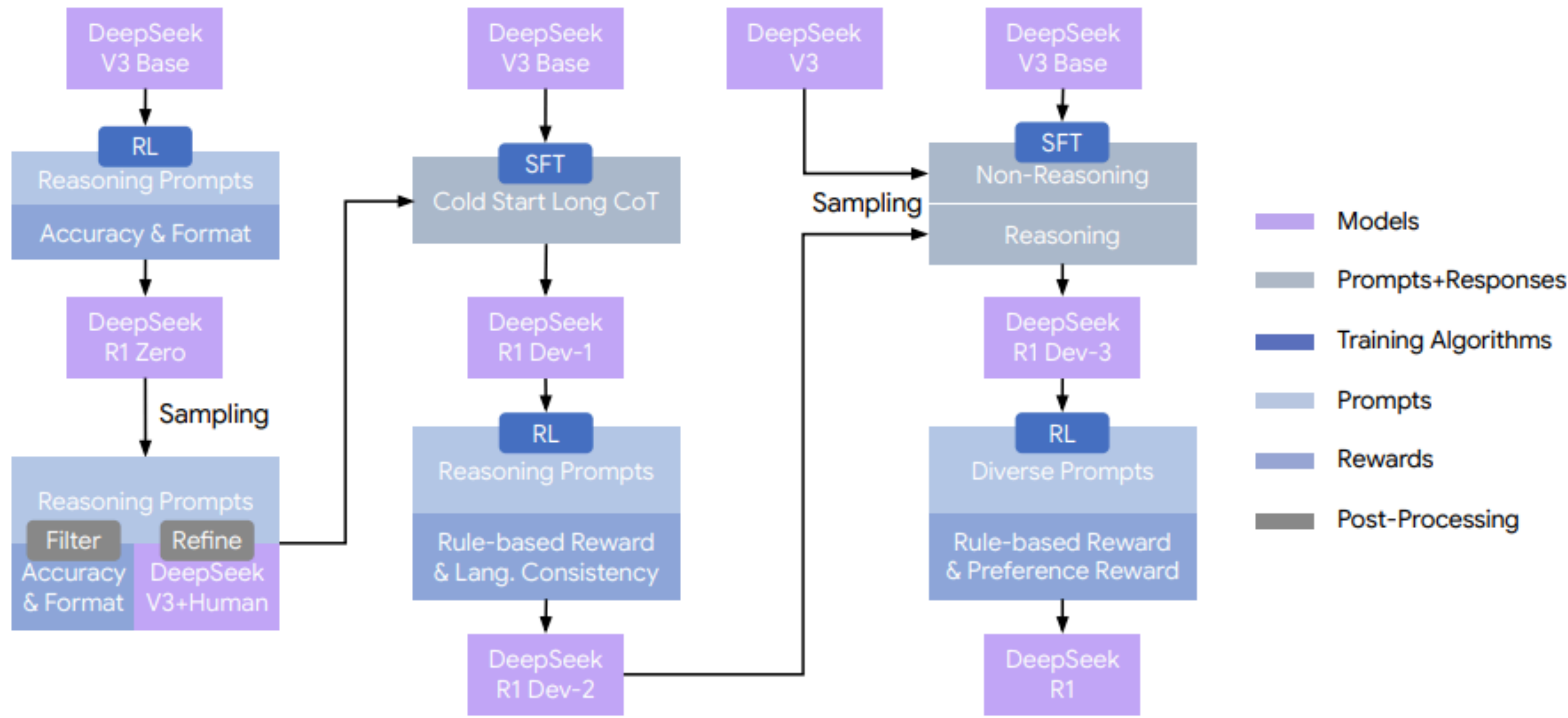


DeepSeek-R1 [DeepSeek 2025] - <https://arxiv.org/pdf/2501.12948>

Deepseek R1

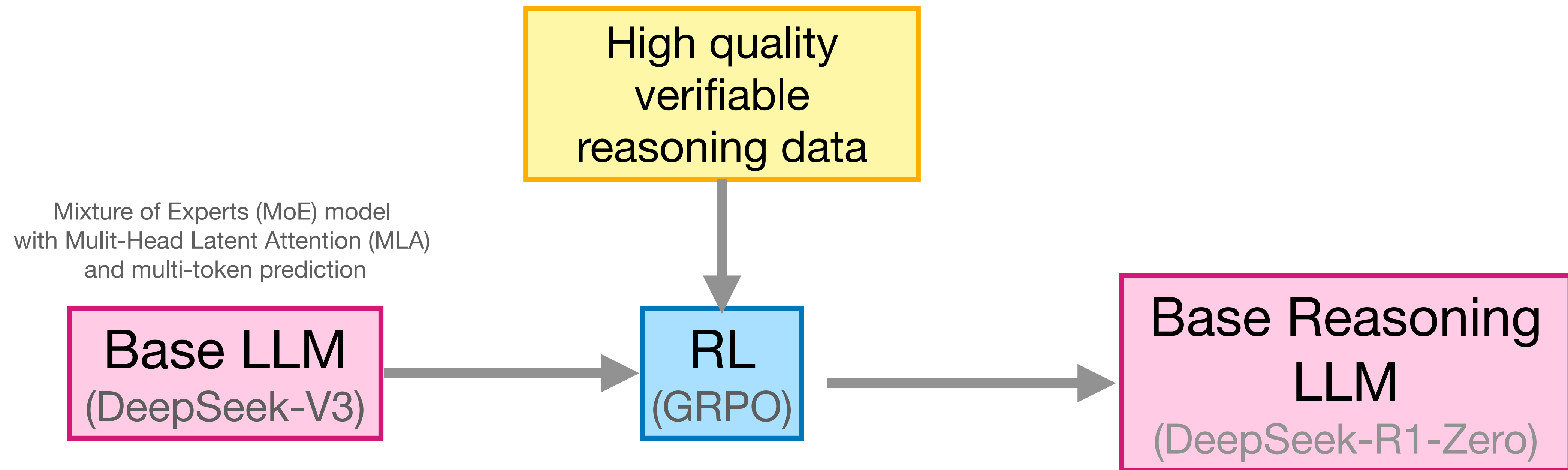
- DeepSeek-V3 - MoE model trained with multi-token prediction
- DeepSeek-R1-Zero - Use RL to improve base model (DeepSeek-V3) for reasoning (e.g. generate “thought” process)
- DeepSeek-R1 - Improved version with cold-start, supervised fine-tuning on generated / filtered data.

Deepseek R1



Deepseek R1-Zero

Training recipe



DeepSeek R1-Zero

Training recipe

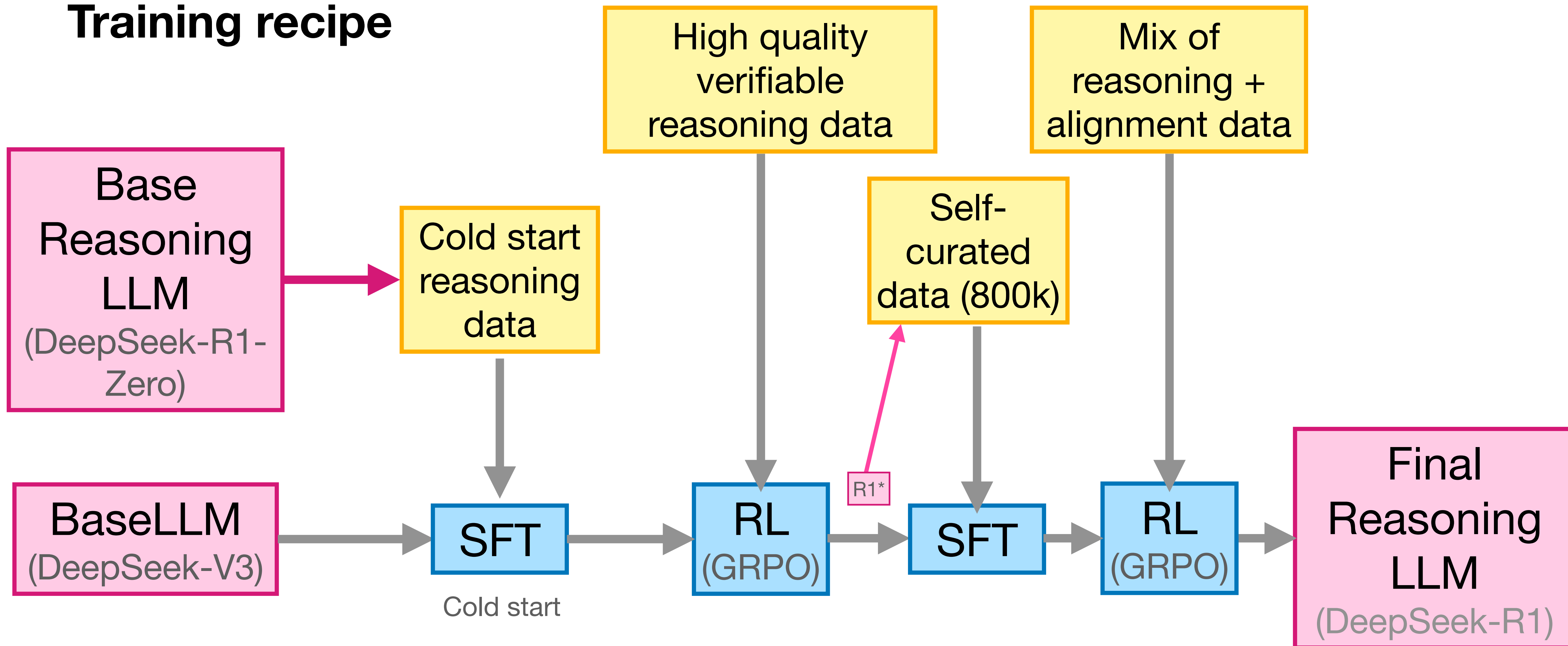
- Train LLM to explicitly generate reasoning process

A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within `<think>` `</think>` and `<answer>` `</answer>` tags, respectively, i.e., `<think> reasoning process here </think>` `<answer> answer here </answer>`. User: **prompt**. Assistant:

- **Accuracy reward** for whether response is correct and **format reward** to provide details of thinking process
- Additional **language consistency reward** is added when training DeepSeek-R1

DeepSeek R1

Training recipe



Reinforcement Learning Review

Some slides adapted from Graham Neubig and others

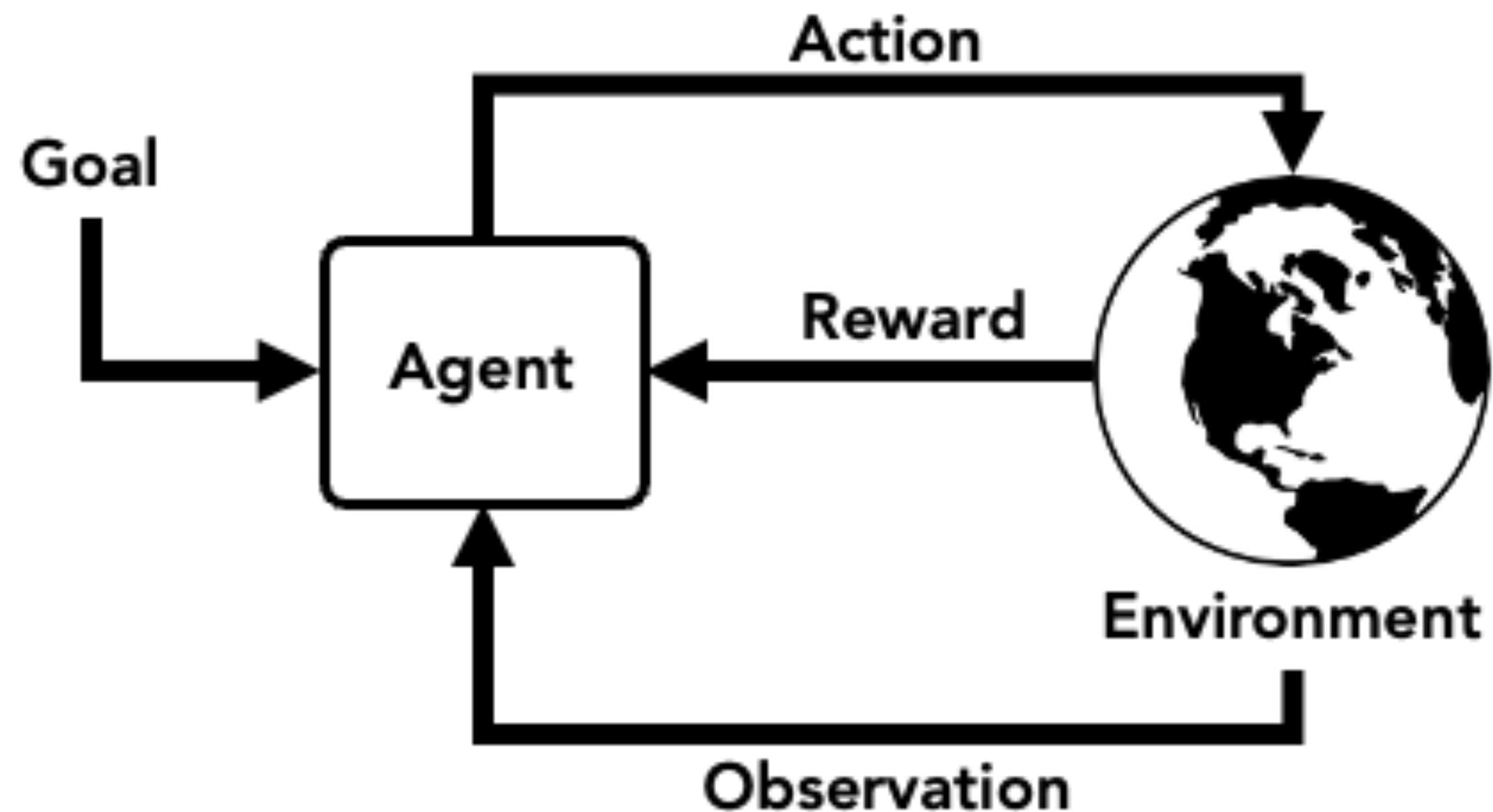
What is reinforcement learning?

Learning where we have an

- Environment X
 - Ability to make actions A
 - Get delayed reward R
- Environment provides feedback
 - No examples of optimal policy

For text generation, we use RL when there is a **sequence-level evaluation metric** that is difficult to optimize without first generating the whole sentence.

Can also be used for longer dialog where reward is at the end (e.g. customer achieved their goal).



Reinforcement learning

Determine policy to **maximize expected accumulated reward**.

Typically modelled as POMDP (sequence of states with partial observations)

- **Actions:** What token to output?
- **Policy:** What action(s) to take given sequence of observations and actions?
 - Policy models the probability of action given state
 - For text generation, what sequence of tokens y to generate given input x
tokens: $\pi(a, s) = P(y | x)$
- **Reward:**
 - RLHF: Provided by reward model trained on human preference
 - RLVR: Provided by verifiable reward / reward

Reward in RL

- For an action at time step t , state s_t and action a_t , get a reward $r_t = R(s_t, a_t)$
- Goal is to maximize **total reward** over time $\sum_{t=0}^{\infty} r_t$
- But maybe better to get immediate reward rather than wait. So maximize **discounted reward** $\sum_{t=0}^{\infty} \gamma^t r_t$
- Everything is stochastic
 - **expected discounted reward** for a policy π

$$\mathbb{E}_{\pi} \left[\sum_t^{\infty} \gamma^t r_t \right]$$

Reward in RL

- For a given trajectory $\tau = ((s_0, a_0), (s_1, a_1), \dots)$, which is a sequence of states and actions in the world, we can have the following rewards.

Also known as **return**

- Total reward:** $R_{\text{total}}(\tau) = \sum_{t=0}^{\infty} r_t$

- Discounted reward:** $R_{\text{discounted}}(\tau) = \sum_{t=0}^{\infty} \gamma^t r_t$

Algorithms that optimize parameters of the policy directly are called **policy-based methods**

- Find a policy π to maximize **expected discounted reward**:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} = \int_{\tau} P(\tau | \pi) R(\tau) = \left[\sum_t^{\infty} \gamma^t r_t \right]$$

RL for text generation

- **State:** input text ($X = x_1, x_2, \dots$) and text that is generated so far ($y_1, y_2, \dots, y_{t-1}$)
- **Action:** next token to generate y_t
- **Trajectory of actions** (generated text): $Y = y_1, \dots, y_n$
- **Objective:** Maximize $J_{\text{RL}}(X) = R(Y) \log P(Y|X)$
- **Loss:**

$$\ell_{\text{RL}}(X) = -R(Y) \log P(Y|X)$$

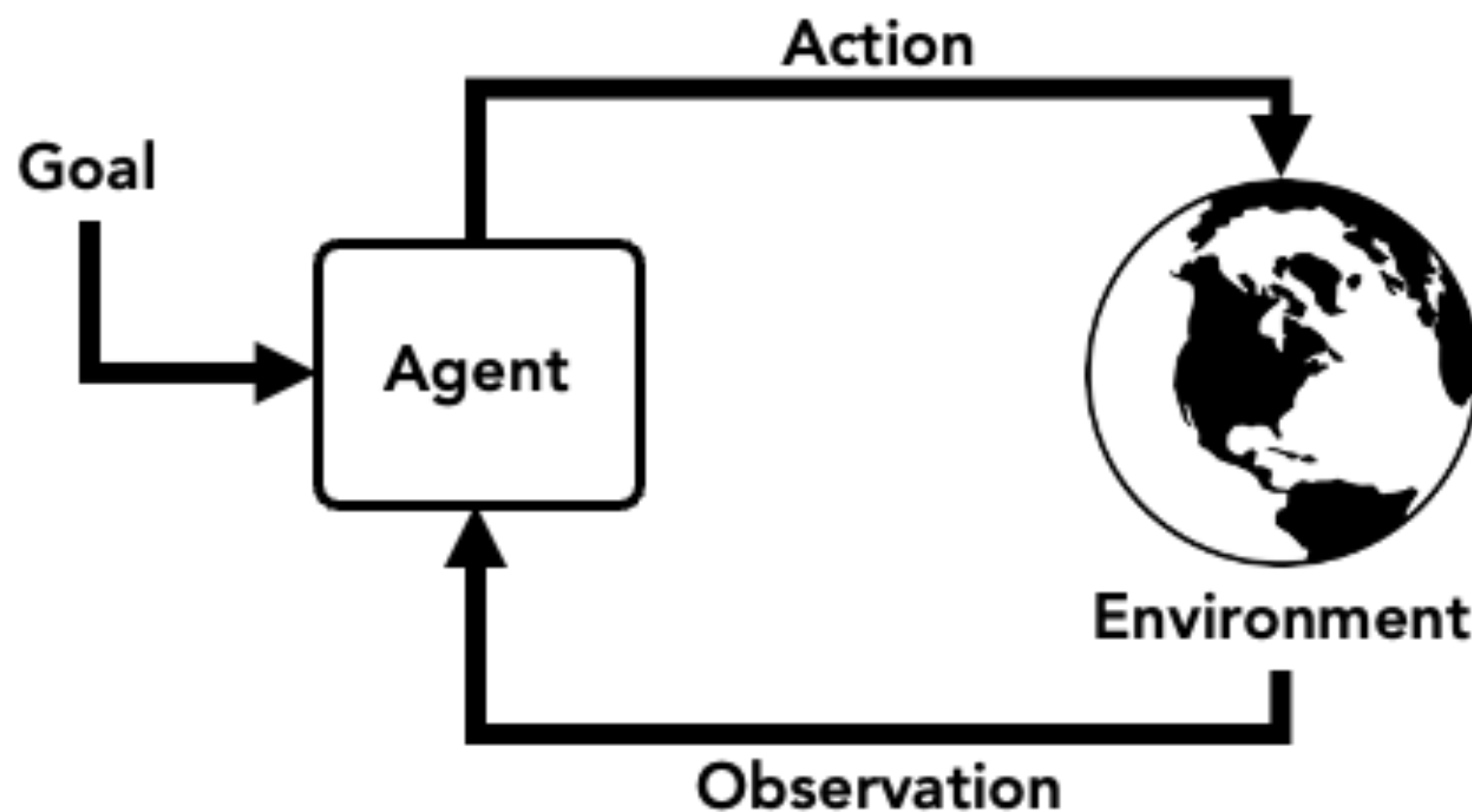
How is this related to supervised learning?

Supervised learning

In normal supervised learning, we do maximum likelihood estimation (MLE)

In the RL literature, this is also known as **imitation learning** where there is an “expert” demonstrating what actions to take at each time step

- Have expert demonstrations (possibly interactive)



$$\ell_{\text{MLE}}(Y|X) = -\log P(Y|X)$$

Training set with
Input text X and output text Y

Examples of imitation learning algorithms:

- Behavior cloning - collect data (expert demonstrations once) and then train
- DAgger - iteratively collect data (as it encounters new states)

Self-training

- **Self-training:** If we start with a trained model, we can sample (or take argmax) to get data points (e.g. “demonstrations”) for training.

$$\hat{Y} \sim P(Y|X) \quad \text{or} \quad \hat{Y} = \arg \max_Y P(Y|X)$$

$$\ell_{\text{self}}(X) = -\log P(\hat{Y}|X)$$

Policy gradient / REINFORCE

- **Self-training:** If we start with a trained model, we can sample (or take argmax) to get data points (e.g. “demonstrations”) for training.

$$\hat{Y} \sim P(Y|X) \quad \text{or} \quad \hat{Y} = \arg \max_Y P(Y|X)$$

$$\ell_{\text{self}}(X) = -\log P(\hat{Y}|X)$$

- **RL:** Add term to scale loss by the reward

RL: Maximize reward

$$J_{\text{REINFORCE}}(X) = R(\hat{Y}) \log P(\hat{Y}|X)$$

$$\ell_{\text{REINFORCE}}(X) = -R(\hat{Y}) \log P(\hat{Y}|X)$$

Optimize policy using stochastic gradient descent

- **But some issues:**
 - step too small, training too slow,
 - too large, training is unstable (too much variability)

Regularization to existing model

- **KL-regularization:**

Improve reward

Keep model similar

$$J_{\text{KL-reg}} = \rho(\hat{Y}, X) R(\hat{Y}) - \beta \text{KL}[P(\cdot|X; \theta_{\text{old}}), P(\cdot|X; \theta)]$$

$$\rho(Y, X) = \frac{P(Y|X; \theta)}{P(Y|X; \theta_{\text{old}})} \quad \text{Ratio function}$$

- **Add regularization term to keep model similar to initial model**
- **Tricky to implement**

Regularization to existing model

- **KL-regularization:**

Improve reward

Keep model similar

$$J_{\text{KL-reg}} = \rho(\hat{Y}, X)R(\hat{Y}) - \beta \text{KL}[P(\cdot|X; \theta_{\text{old}}), P(\cdot|X; \theta)]$$

$$\rho(Y, X) = \frac{P(Y|X; \theta)}{P(Y|X; \theta_{\text{old}})}$$

Ratio function

- **Proximal policy optimization (PPO)**

Prevent jumps

$$J_{\text{PPO}} = \min \left(\underbrace{\rho(\hat{Y}, X)R(\hat{Y})}_{\text{Unclipped objective}}, \underbrace{\text{clip}(\rho(\hat{Y}, X), 1 + \epsilon, 1 - \epsilon)R(\hat{Y})}_{\text{Clipped objective}} \right)$$

Unclipped objective

Clipped objective

Going beyond reward

Hard to compute, so use
neural networks to estimate
these functions
(critic training)

- State **Value** Function
 - How good is the state s_t ?
- State-Action Value Function (**Q function**)
 - How good is a particular action a at state s_t ?
- **Advantage**
 - How much better is taking action a than the average?

$$V_{\pi}(s_t) = \mathbb{E}_{\pi} \left[\sum_{i=t}^{\infty} \gamma^{i-t} r_i \right]$$

$$Q_{\pi}(s_t, a) = \mathbb{E}_{s_{t+1}} \left[R(s_t, a, s_{t+1}) + \gamma [V_{\pi}(s_{t+1})] \right]$$

$$A_{\pi}(s_t, a) = Q_{\pi}(s_t, a) - V_{\pi}(s_t)$$

Use as return function
in optimization:

$$J(X) = R(\hat{Y}) \log P(\hat{Y}|X) \longrightarrow J(X) = A(\hat{Y}) \log P(\hat{Y}|X)$$

PPO with advantage

- **Proximal policy optimization (PPO)**

$$J_{\text{PPO}} = \min \left(\rho(\hat{Y}, X) \underline{\hat{A}(\hat{Y})}, \text{clip}(\rho(\hat{Y}, X), 1 + \epsilon, 1 - \epsilon) \underline{\hat{A}(\hat{Y})} \right)$$

$$\rho(Y, X) = \frac{P(Y|X; \theta)}{P(Y|X; \theta_{\text{old}})} \quad \text{Ratio function}$$

On-policy, actor-critic algorithm

Actor: Policy network - determines what action to take

Critic: Value network - evaluates action by estimating the value function
(so we can estimate the advantage function) $A(\hat{Y})$

RLHF with general policy gradient method

- Initialize the policy π_{ϕ}^{RL} to π^{SFT} , the policy output from SFT

- Loop for many steps

- Initialize a new empty dataset $D_{\pi_{\phi}^{\text{RL}}}$

- Loop for many steps

- Sample a random prompt x from D_{RL}

- Generate a response y from the policy π_{ϕ}^{RL}

- Calculate the reward signal $r_{\theta}(x, y)$ from r_{θ}

- Add the triple $(x, y, r_{\theta}(x, y))$ to $D_{\pi_{\phi}^{\text{RL}}}$
- $$\text{objective}(\phi) = E_{(x,y) \sim D_{\pi_{\phi}^{\text{RL}}}} \left[r_{\theta}(x, y) - \beta \log \left(\frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi^{\text{SFT}}(y|x)} \right) \right]$$

- Use policy gradient method to increase objective

RLHF with PPO

Algorithm 1 PPO

- 1: Input: initial policy parameters θ_0 , initial value function parameters ϕ_0 .
- 2: **for** $n = 0, 1, 2, \dots$ **do**
- 3: Collect a set of trajectories $\mathcal{D}_n = \{\tau_i\}$ by executing policy $\pi(\theta_n)$ within the environment.
- 4: Compute rewards-to-go $\hat{R}_t$.
- 5: Compute advantage estimates, $\hat{A}_t$ (using any advantage estimation method) based on the current value function V_{ϕ_n} .
- 6: Update the policy by maximizing the PPO-penalty/clip/ptx objective:

$$\theta_{n+1} = \arg \max_{\theta} \mathcal{L}_{\text{ppo-clip}}(\theta_n).$$

- 7: Update the value function by regression on mean-squared error:

$$\phi_{n+1} = \arg \min_{\phi} \mathcal{L}_{\text{critic}}(\phi_n).$$

- 8: **end for**
-

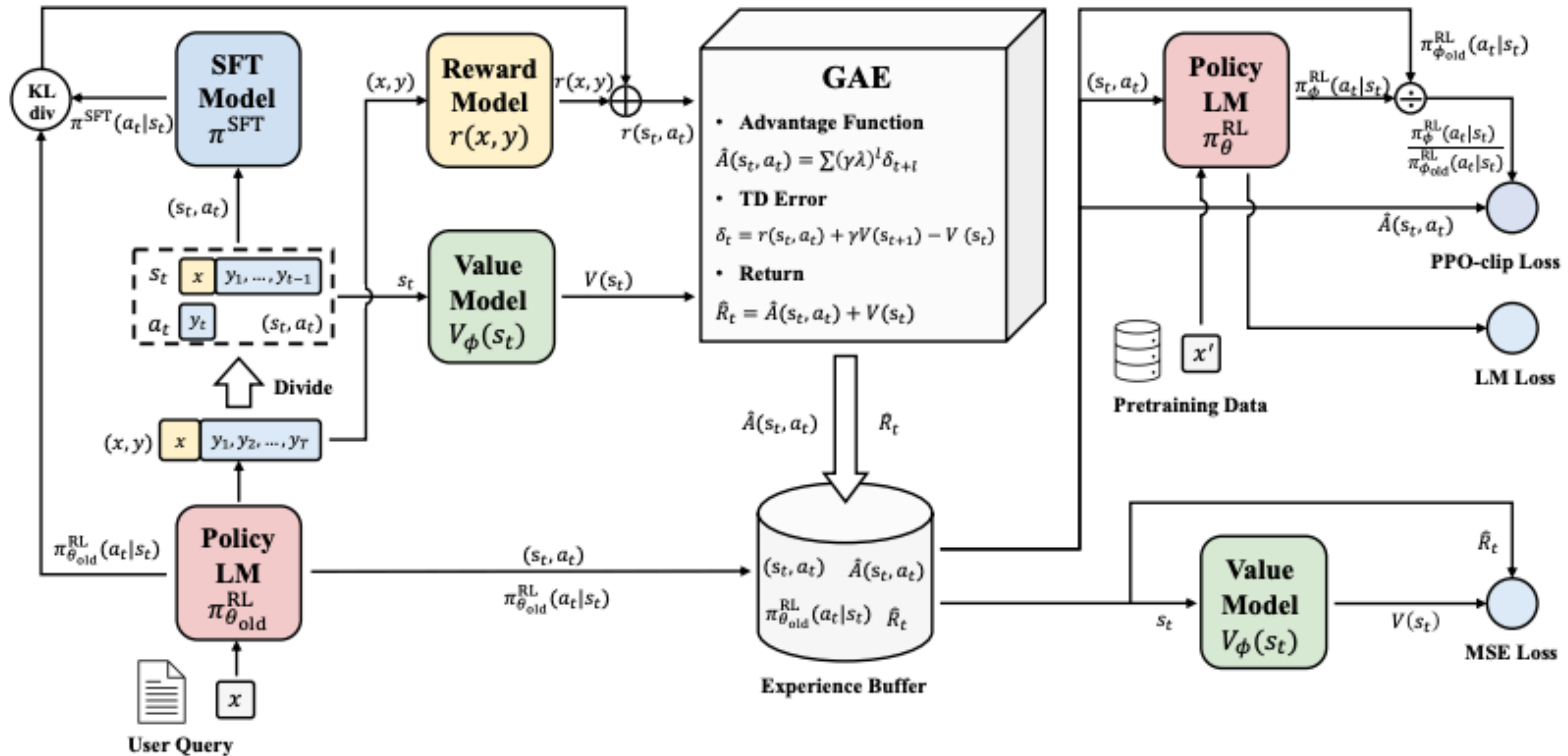


Figure 1: PPO workflow, depicting the sequential steps in the algorithm's execution. The process begins with sampling from the environment, followed by the application of GAE for improved advantage approximation. The diagram then illustrates the computation of various loss functions employed in PPO, signifying the iterative nature of the learning process and the policy updates derived from these losses.

Group Relative Policy Optimization (GRPO)

- Replace advantage function estimated with neural reward model with advantage function that is computed from rule-based reward model
- Compute advantage as how much newer policy improves over older policy

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)]$$

$$\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left\{ \min \left[\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})} \hat{A}_{i,t}, \text{clip} \left(\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{i,t} \right] - \beta \mathbb{D}_{KL} [\pi_{\theta} || \pi_{ref}] \right\}.$$

$$\mathbb{D}_{KL} (\pi_{\theta} || \pi_{ref}) = \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - \log \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - 1,$$

Group Relative Policy Optimization (GRPO)

- Outcome supervision

$$\hat{A}_{i,t} = \tilde{r}_i = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}$$

$$\mathbf{r} = \{r_1, \dots, r_G\}$$

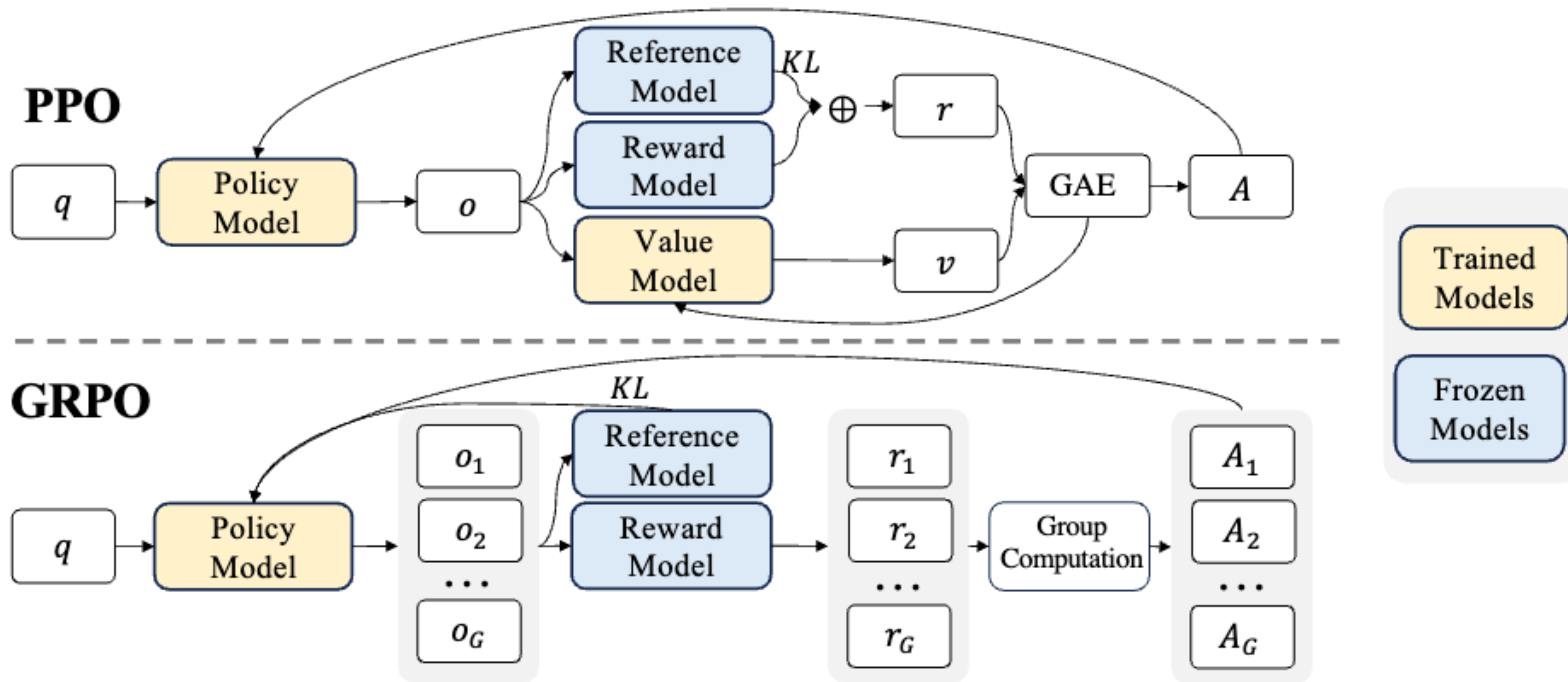
- Process supervision

$$\hat{A}_{i,t} = \sum_{index(j) \geq t} \tilde{r}_i^{index(j)}$$

$$\tilde{r}_i^{index(j)} = \frac{r_i^{index(j)} - \text{mean}(\mathbf{R})}{\text{std}(\mathbf{R})}$$

$$\mathbf{r} = \{ \{ r_1^{index(1)}, \dots, r_1^{index(K_1)} \}, \dots, \{ r_G^{index(1)}, \dots, r_G^{index(K_G)} \} \}$$

Group Relative Policy Optimization (GRPO)



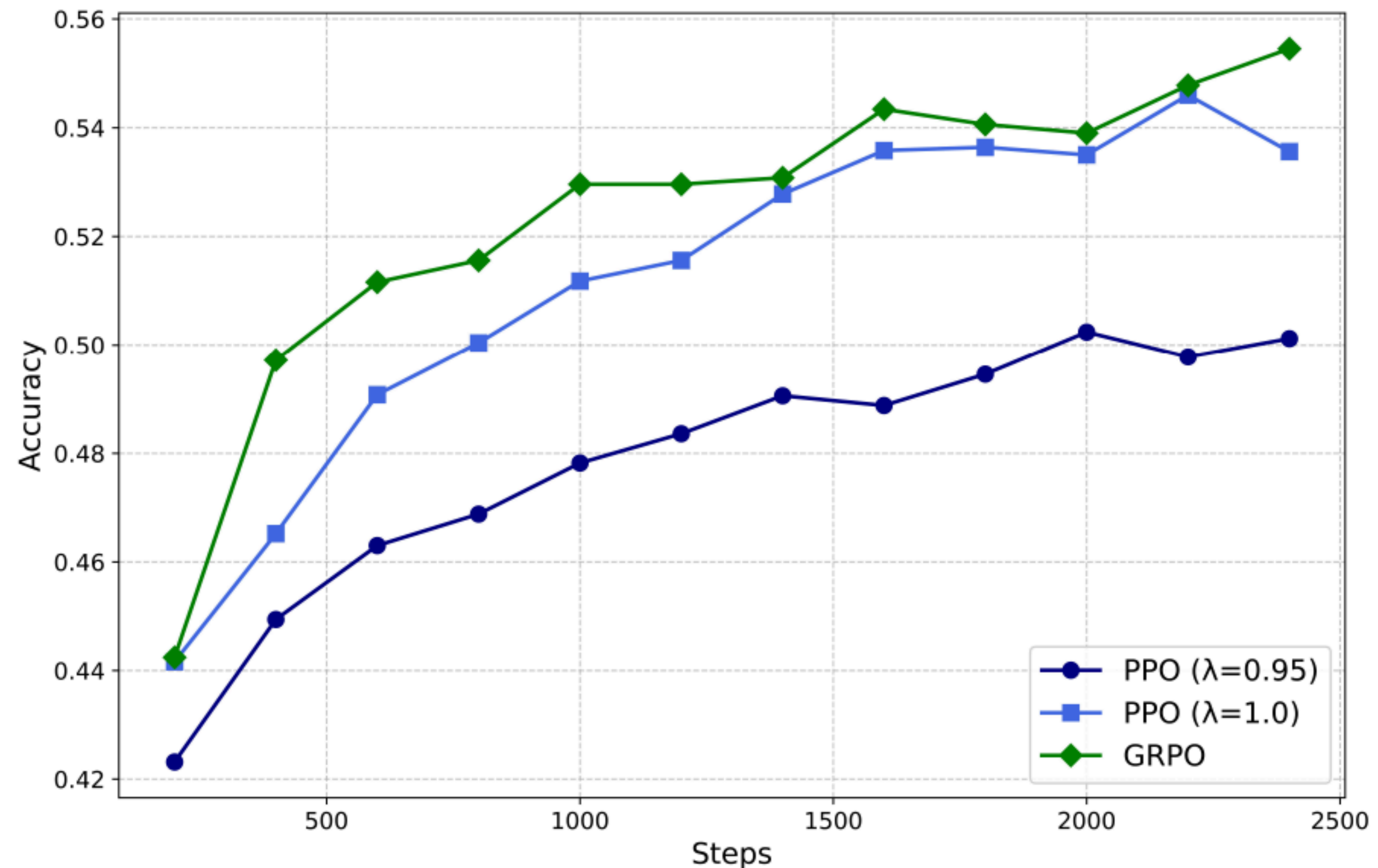
Eliminate value model

References for PPO and GRPO

- PPO for LLMs: <https://cameronrwolfe.substack.com/p/ppo-llm>
- GRPO: <https://cameronrwolfe.substack.com/p/grpo>
- GRPO Tricks: <https://cameronrwolfe.substack.com/p/grpo-tricks>
- Online vs Offlice: <https://cameronrwolfe.substack.com/p/online-rl>

Comparison of PPO and GRPO

MATH task



DeepSeek-R1 performance

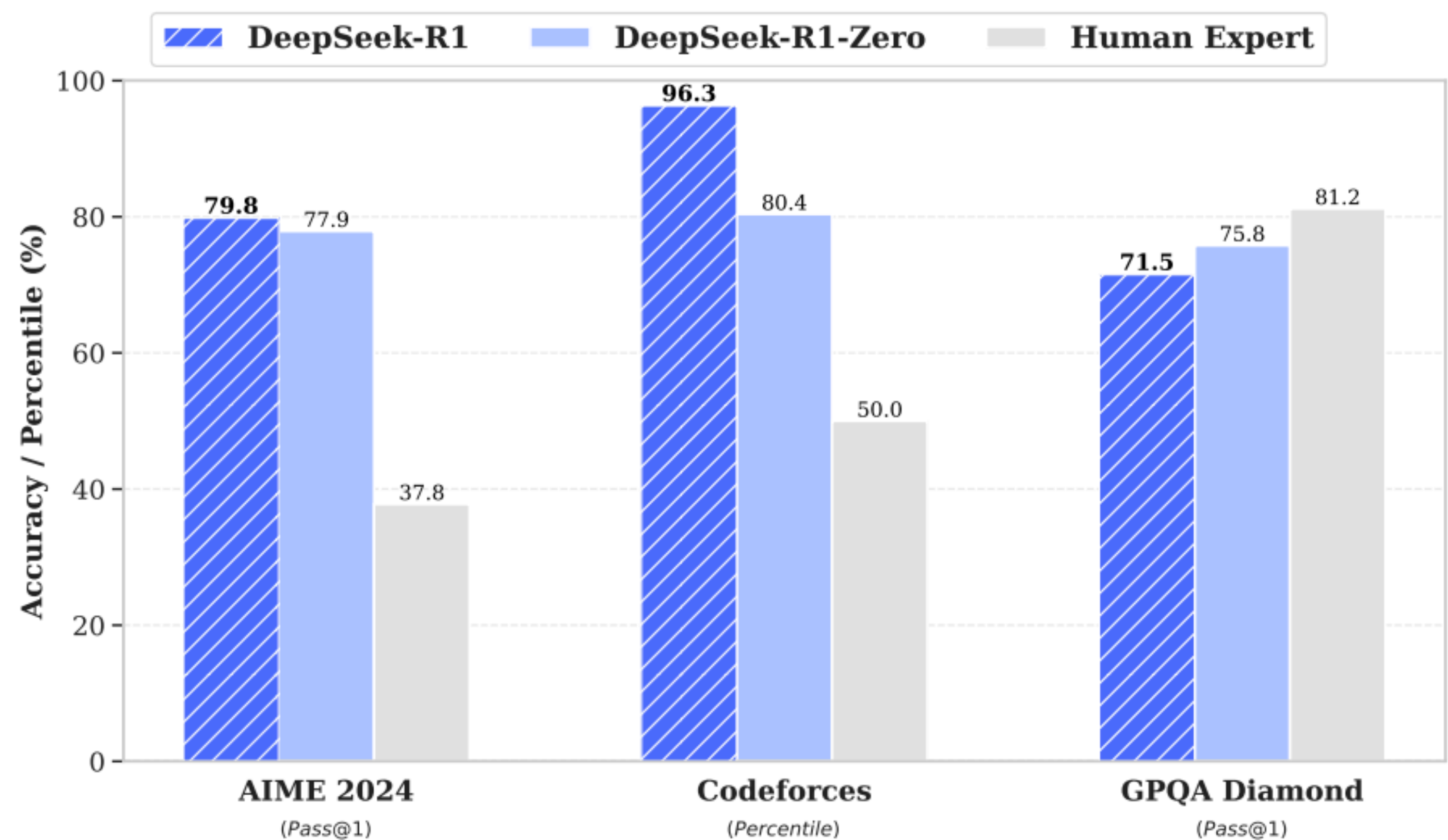
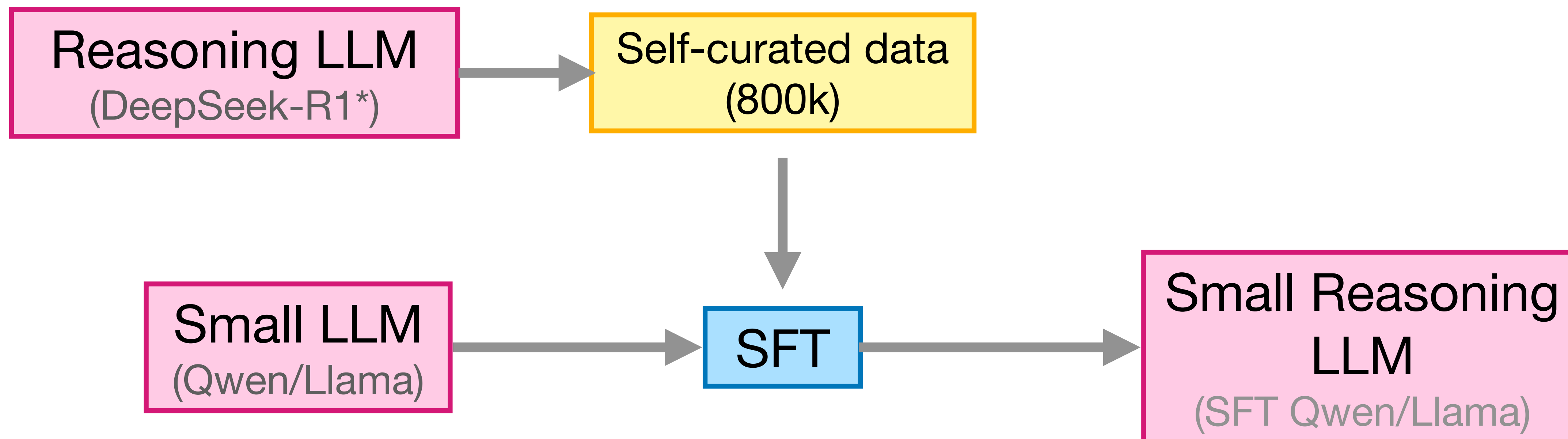


Figure 10 | The benchmark performance of DeepSeek-R1 and DeepSeek-R1-Zero is compared with human scores across different datasets. For AIME and Codeforces, the human scores represent the average performance of all human competitors. In the case of GPQA, the human score corresponds to Ph.D.-level individuals who had access to the web for answering the questions.

Distillation of DeepSeek-R1

Training recipe

Generated / Filtered using DeepSeek-R1*
Reasoning (600K)
General (200K)



DeepSeek-R1 performance

Distilled models

Table 15 | Comparison of DeepSeek-R1 distilled models and other comparable models on reasoning-related benchmarks. Numbers in bold denote the performance is statistically significant (t-test with $p < 0.01$).

Model	AIME 2024		MATH	GPQA	LiveCode	CodeForces
	pass@1	cons@64		Diamond	Bench	
			pass@1	pass@1	pass@1	rating
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	72.6	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	86.7	94.5	65.2	57.5	1633

Does RL increase the reasoning capacity?

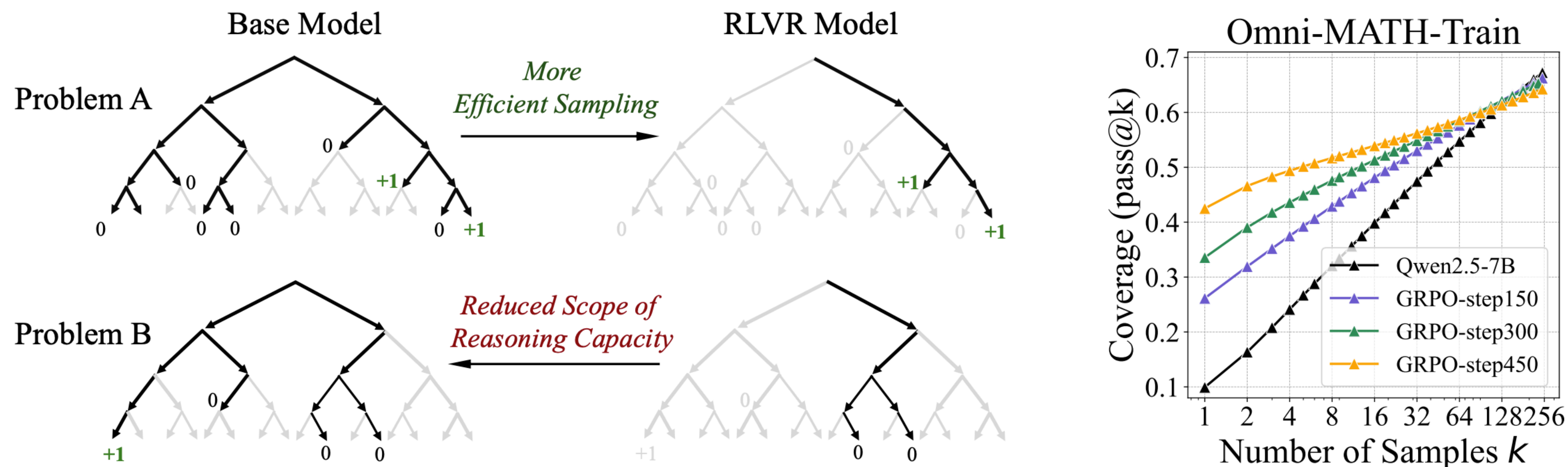
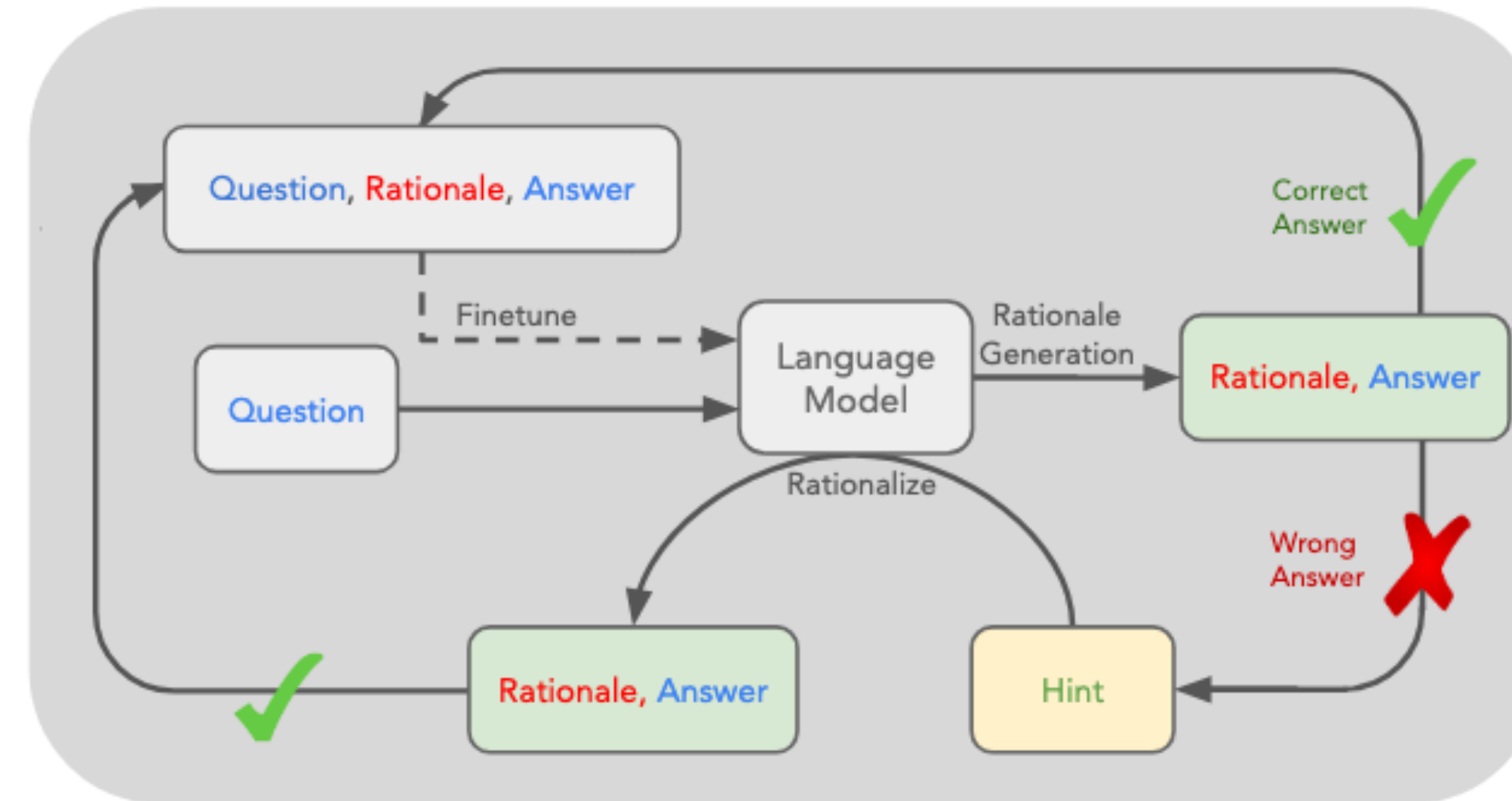


Figure 1: **(Left)** The effect of current RLVR on LLM’s reasoning ability. Search trees are generated by repeated sampling from the base and RLVR-trained models for a given problem. Grey indicates paths that are unlikely to be sampled by the model, while **black** indicates paths that are likely to be sampled. **Green** indicates correct paths, which has positive rewards. Our key finding is that all reasoning paths in the RLVR model are already present in the base model. For certain problems like Problem A, RLVR training biases the distribution toward rewarded paths, improving sampling efficiency. However, this comes at the cost of reduced scope of reasoning capacity: For other problems like Problem B, the base model contains the correct path, whereas that of the RLVR model does not. **(Right)** As RLVR training progresses, the average performance (*i.e.*, pass@1) improves, but the coverage of solvable problems (*i.e.*, pass@256) decreases, indicating a reduction in LLM’s reasoning boundary.

Self-training

STaR: Self-Taught Reasoner

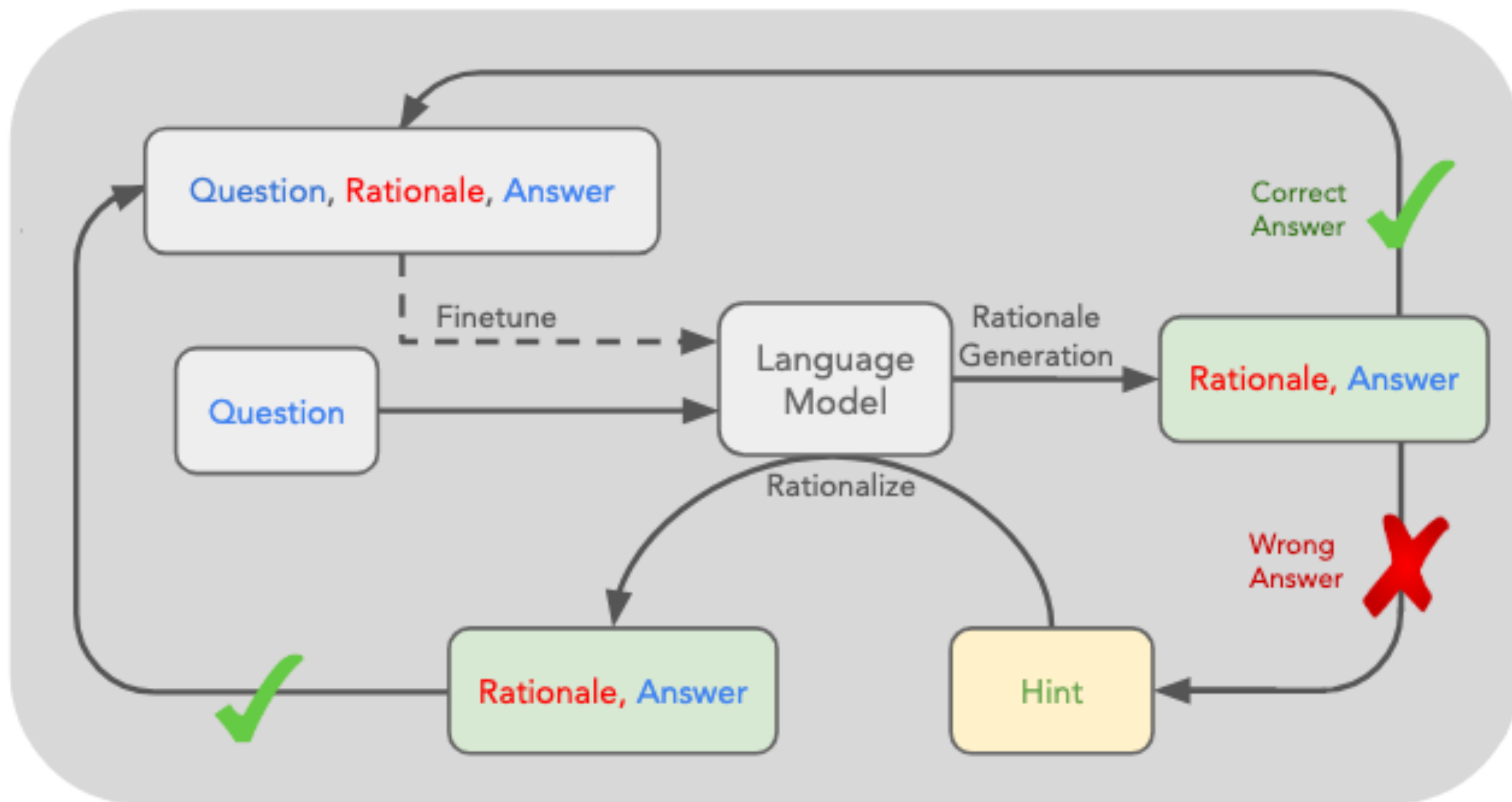
- Start with a pretrained LLM
- Assumes dataset with **questions** and **answers**.
- Generate **rationales**
- Fine-tune LLM on **correctly** generated answer with rationale
- If it cannot generate the correct answer, provide **hint** of the correct answer



Q: What can be used to carry a small dog?
Answer Choices:
(a) swimming pool
(b) basket
(c) dog show
(d) backyard
(e) own home
A: The answer must be something that can be used to carry a small dog. Baskets are designed to hold things. Therefore, the answer is basket (b).

Q: Where do you put your grapes just before checking out?
Answer Choices:
(a) mouth
(b) grocery cart (CORRECT)
(c) super market
(d) fruit basket
(e) fruit market

STaR: Self-Taught Reasoner



Q: What can be used
to carry a small dog?

Answer Choices:

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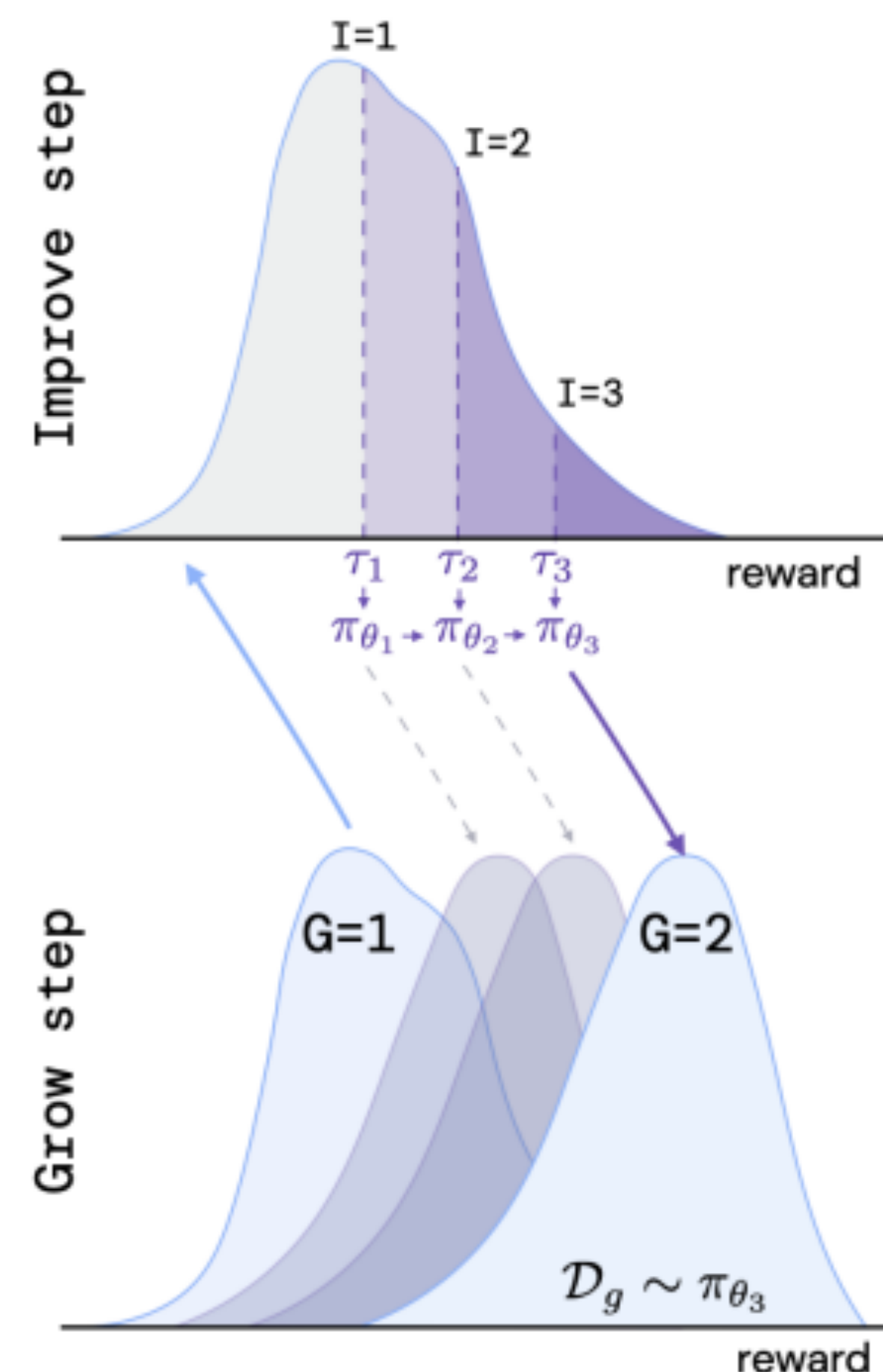
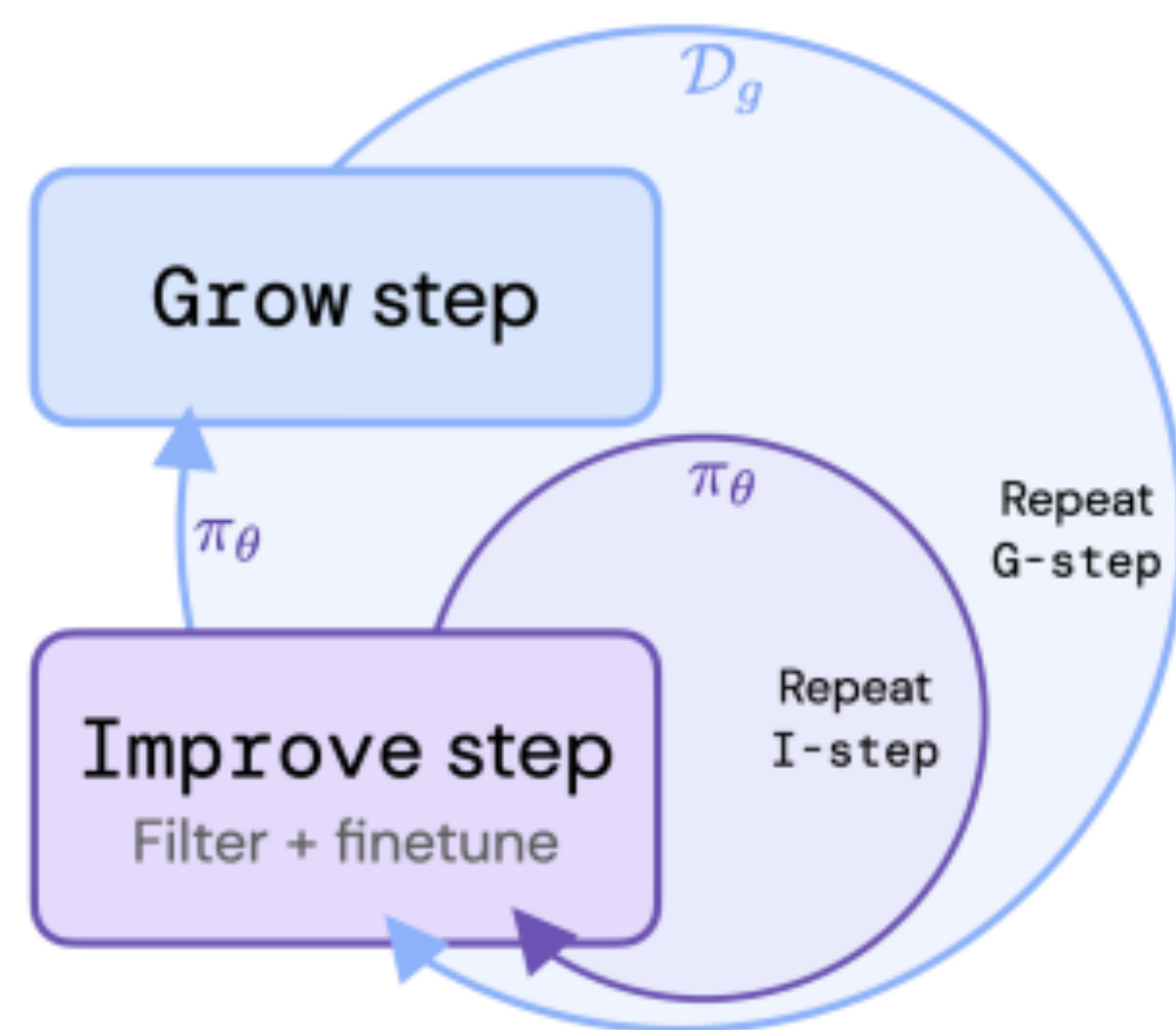
Algorithm 1 STaR

Input M : a pretrained LLM; dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^D$ (w/ few-shot prompts)

- 1: $M_0 \leftarrow M$ # Copy the original model
 - 2: **for** n **in** $1 \dots N$ **do** # Outer loop
 - 3: $(\hat{r}_i, \hat{y}_i) \leftarrow M_{n-1}(x_i) \quad \forall i \in [1, D]$ # Perform rationale generation
 - 4: $(\hat{r}_i^{\text{rat}}, \hat{y}_i^{\text{rat}}) \leftarrow M_{n-1}(\text{add_hint}(x_i, y_i)) \quad \forall i \in [1, D]$ # Perform rationalization
 - 5: $\mathcal{D}_n \leftarrow \{(x_i, \hat{r}_i, y_i) \mid i \in [1, D] \wedge \hat{y}_i = y_i\}$ # Filter rationales using ground truth answers
 - 6: $\mathcal{D}_n^{\text{rat}} \leftarrow \{(x_i, \hat{r}_i^{\text{rat}}, y_i) \mid i \in [1, D] \wedge \hat{y}_i \neq y_i \wedge \hat{y}_i^{\text{rat}} = y_i\}$ # Filter rationalized rationales
 - 7: $M_n \leftarrow \text{train}(M, \mathcal{D}_n \cup \mathcal{D}_n^{\text{rat}})$ # Finetune the original model on correct solutions - inner loop
 - 8: **end for**
-

ReST: Reinforced Self-Training

- **Grow** - generate dataset for training
- **Improve** - use generated dataset to fine-tune the policy



Better policy is trained on better datasets

Dataset is iteratively filtered to be better (with higher reward thresholds)

ReST: Reinforced Self-Training

Algorithm 1: ReST algorithm. *ReST* is a growing-batch RL algorithm. Given an initial policy of reasonable quality (for example, pre-trained using BC) iteratively applies Grow and Improve steps to update the policy. Here F is a filtering function, and $\mathcal{L}$ is an loss function.

Input: $\mathcal{D}$: Dataset, $\mathcal{D}_{eval}$: Evaluation dataset, $\mathcal{L}(\mathbf{x}, \mathbf{y}; \theta)$: loss, $R(\mathbf{x}, \mathbf{y})$: reward model, G : number of grow steps, I : number of improve steps, N : number of samples per context

Train π_θ on $\mathcal{D}$ using loss $\mathcal{L}$.

for $g = 1$ to G **do**

// Grow

 Generate dataset $\mathcal{D}_g$ by sampling: $\mathcal{D}_g = \{ (\mathbf{x}^i, \mathbf{y}^i) |_{i=1}^{N_g} \text{ s.t. } \mathbf{x}^i \sim \mathcal{D}, \mathbf{y}^i \sim \pi_\theta(\mathbf{y}|\mathbf{x}^i) \} \cup \mathcal{D}$.

 Annotate $\mathcal{D}_g$ with the reward model $R(\mathbf{x}, \mathbf{y})$.

for $i = 1$ to I **do**

// Improve

 Choose threshold s.t. $\tau_1 > V_{\pi_\theta}$ for $V_{\pi_\theta} = \mathbb{E}_{\mathcal{D}_g} [R(\mathbf{x}, \mathbf{y})]$ and $\tau_{i+1} > \tau_i$.

while *reward improves on $\mathcal{D}_{eval}$* **do**

 | Optimise θ on objective: $J(\theta) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}_g} [F(\mathbf{x}, \mathbf{y}; \tau_i) \mathcal{L}(\mathbf{x}, \mathbf{y}; \theta)]$

end

end

end

Output: Policy π_θ

ReST: Reinforced Self-Training

Improved performance on MT with multiple improve steps

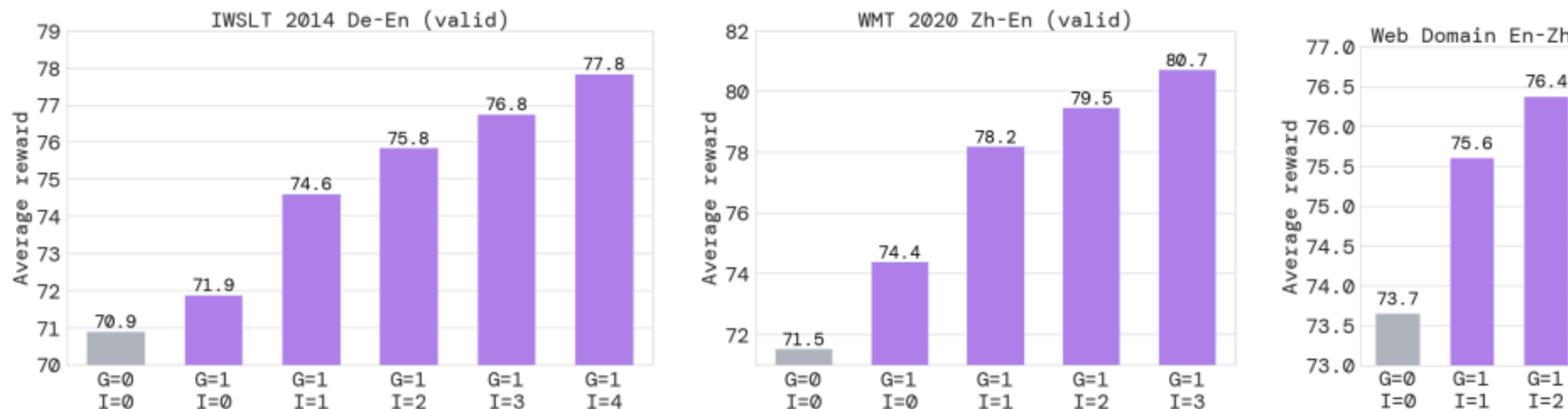


Figure 3 | **ReST with multiple Improve steps.** Average reward model scores on IWSLT 2014 De-En, WMT 2020 Zh-En, and Web Domain En-Zh validation sets. On each dataset, we report results with BC ($G = 0$, $I = 0$) and ReST with a single Grow step and several Improve steps with an increasing reward threshold. Each Improve step increases the reward model score in all three validation datasets. We found the suitable number of Improve steps to be a dataset-dependent hyperparameter.

ReST: Reinforced Self-Training

Improved performance on MT with two grow steps

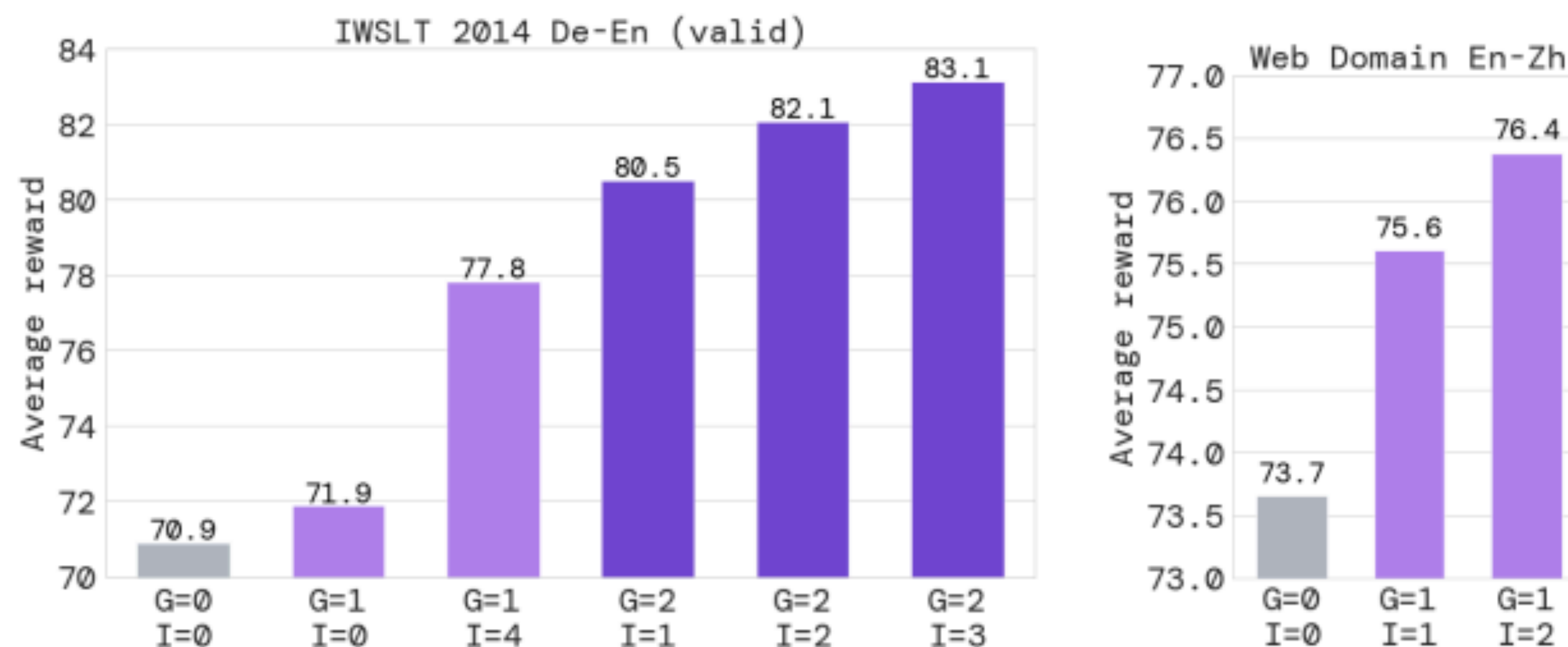


Figure 4 | **ReST with two Grow steps.** The second Grow step with subsequent Improve steps improves the performance by 5.3 points on IWSLT 2014 De-En and 0.8 points on Web Domain En-Zh task over the first Grow step.

ReSTEM: EM for Reinforced Self-Training

- Expectation maximization for reinforced self-training - simpler version of ReST
 - Generate (E-step) - Generate data by sampling + annotate with reward
 - Improve (M-step) - Optimize parameters with generated data
- Differences from ReST
 - Focus on problems with binary rewards (0/1) vs bounded real-value rewards

ReSTEM: EM for Reinforced Self-Training

Algorithm 1: ReST (Expectation-Maximization). Given a initial policy (e.g., pre-trained LM), ReST^{EM} iteratively applies Generate and Improve steps to update the policy.

Input: $\mathcal{D}$: Training dataset, $\mathcal{D}_{val}$: Validation dataset, $\mathcal{L}(\mathbf{x}, \mathbf{y}; \theta)$: loss, $r(\mathbf{x}, \mathbf{y})$: Non-negative reward function, I : number of iterations, N : number of samples per context

for $i = 1$ **to** I **do**

// Generate (E-step)

 Generate dataset $\mathcal{D}_i$ by sampling: $\mathcal{D}_i = \{ (\mathbf{x}^j, \mathbf{y}^j) |_{j=1}^N \text{ s.t. } \mathbf{x}^j \sim \mathcal{D}, \mathbf{y}^j \sim p_\theta(\mathbf{y}|\mathbf{x}^j) \}$

 Annotate $\mathcal{D}_i$ with the reward $r(\mathbf{x}, \mathbf{y})$.

// Improve (M-step)

while *reward improves on $\mathcal{D}_{val}$* **do**

 | Optimise θ to maximize objective: $J(\theta) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}_i} [r(\mathbf{x}, \mathbf{y}) \log p_\theta(\mathbf{y}|\mathbf{x})]$

end

end

Output: Policy p_θ

ReST^{EM}: EM for Reinforced Self-Training

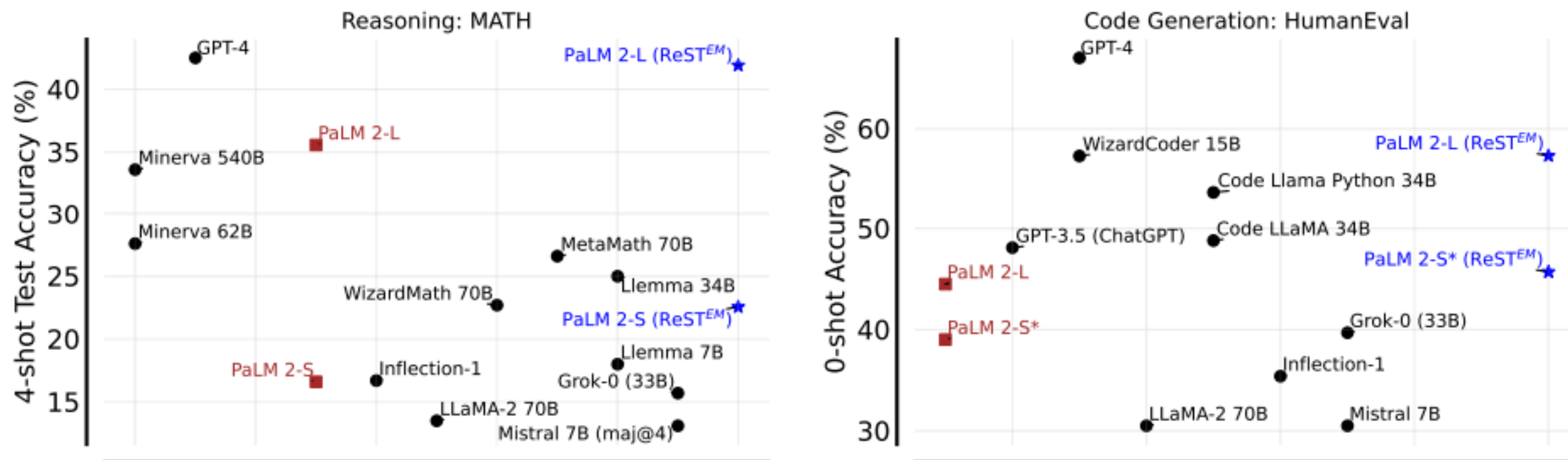


Figure 1 | Self-training with ReST^{EM} substantially improves test performance of PaLM 2 models on two challenging benchmarks: MATH and HumanEval. Results for other models are shown for general progress on these tasks and are typically not comparable due to difference in model scales. GPT-4 results are taken from [Bubeck et al. \(2023\)](#). The x-axis approximately denotes release time (not to scale).

ReST-MCTS*: LLM Self-Training via Process Reward Guided Tree Search

- Includes training of process reward model
- Iteratively
 - Generate new traces / filter for high reward traces
 - Improve policy model
 - Collect process ward
 - Train process reward model

ReST-MCTS*: LLM Self-Training via Process Reward Guided Tree Search

Algorithm 1: Mutual self-training ReST-MCTS* for value model and policy model.

Input: base LLM π , original dataset for policy model D_{S_0} , original dataset for value model D_0 , new problem set D_G , number of solutions N , j -th solution A_j , correct solution a^* , value model V_θ , weighted value function w , quality value function v , number of iterations T .

```
1:  $\pi_{S_0} \leftarrow \text{SFT}(\pi, D_{S_0})$  // fine-tune generator
2:  $D_{V_0} \leftarrow \text{generate\_value\_data}(D_0, w, v)$  // initialize train set for value model
3:  $V_0 \leftarrow \text{train\_value\_model}(V_\theta, D_{V_0})$  // initialize value model
4: for  $i = 1$  to  $T$  do
5:    $D_{G_i} \leftarrow \text{generate\_policy\_data}(\pi_{S_{i-1}}, V_{i-1} \text{ guided MCTS}^*, D_G, N)$  // generate
     synthetic data for policy model
6:   for  $j = 1$  to  $N$  do
7:      $D_{G_i}(A_j=a^*) \leftarrow \text{label\_correctness}(D_{G_i})$  // match and select correct solutions
8:   end for
9:    $\pi_{S_i} \leftarrow \text{SFT}(\pi_{S_{i-1}}, D_{G_i}(A_j=a^*)|_{j=1}^N)$  // self-training policy model
10:   $D_{V_i} \leftarrow \text{extract\_value\_data}(D_{G_i})$  // collect process reward and extract value data
11:   $V_i \leftarrow \text{train\_value\_model}(V_{i-1}, D_{V_i})$  // self-training value model
12: end for
Output:  $\pi_{S_T}, V_T$ 
```

ReST-MCTS*: LLM Self-Training via Process Reward Guided Tree Search

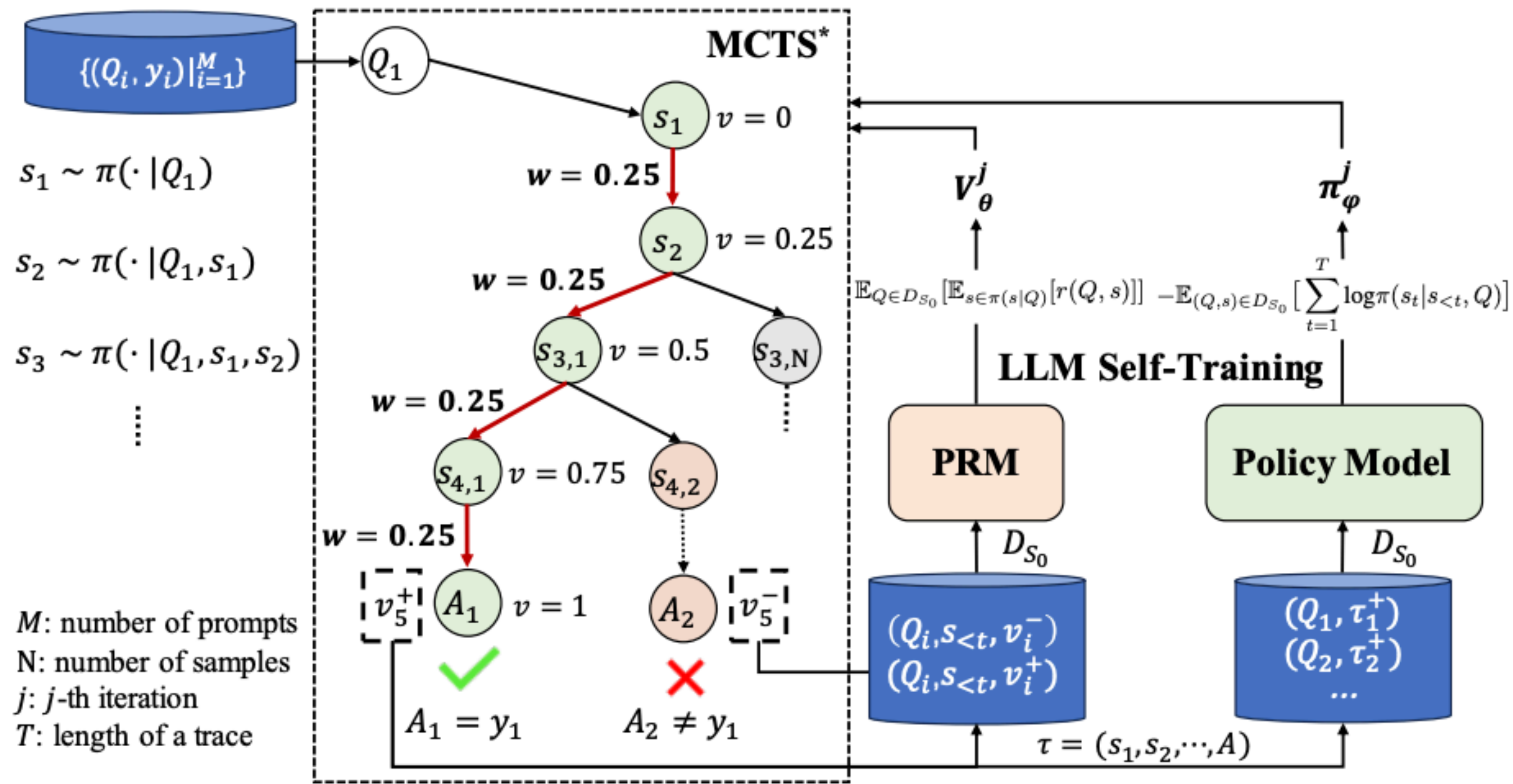


Figure 1: The left part presents the process of inferring process rewards and how we conduct process reward guide tree-search. The right part denotes the self-training of both the process reward model and the policy model.

ReST-MCTS*: LLM Self-Training via Process Reward Guided Tree Search

Table 1: Key differences between existing self-improvement methods and our approach. Train refers to whether to train a reward model.

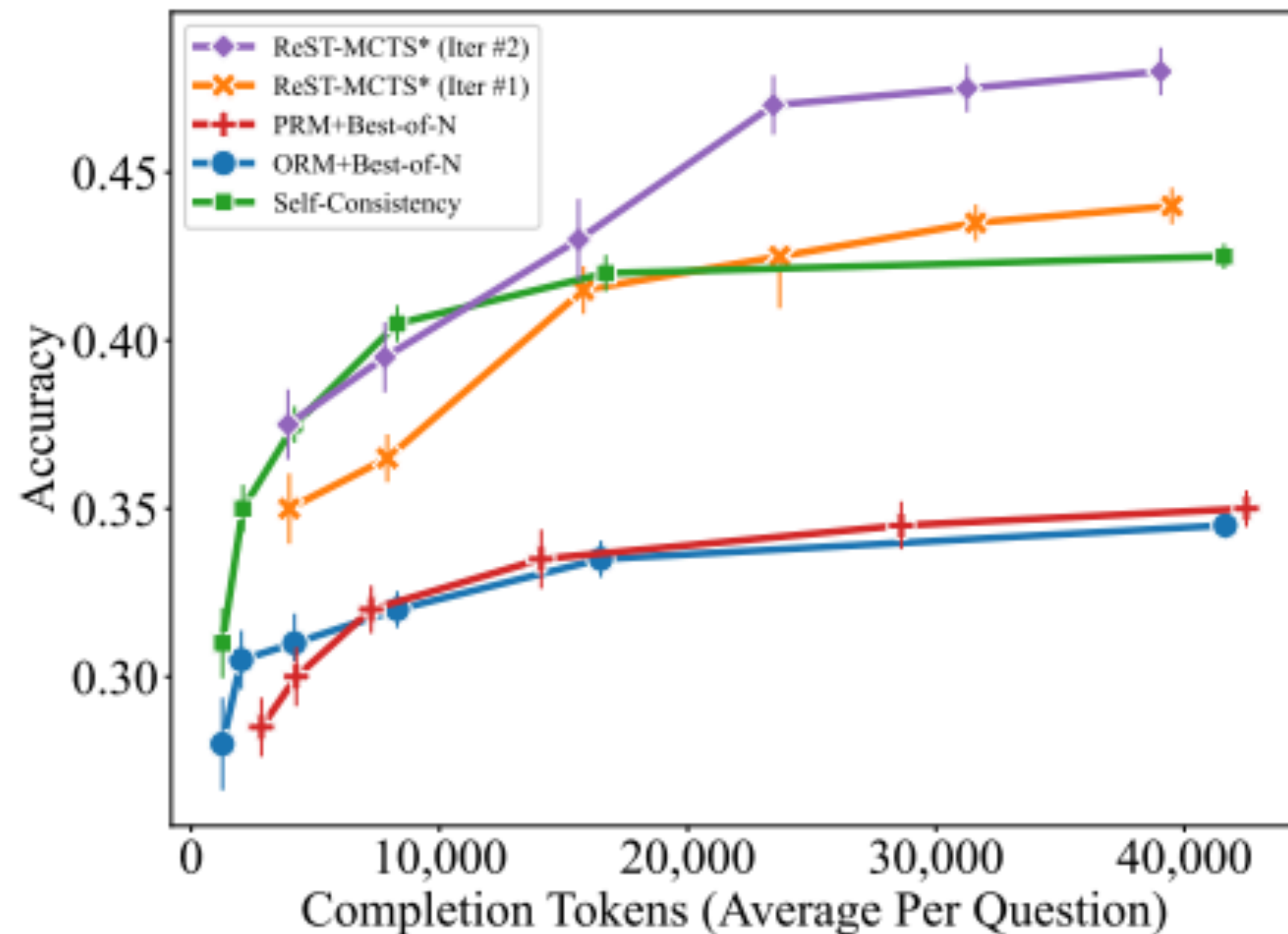
Method	Reasoning Policy	Reward Guidance	
	Method	Value Label	Train
STaR [4] ReST ^{EM} [6] / RFT [5] / RPO [11]	CoT+Reflexion	Final outcome reward annotated by ground-truth answer	✗
	CoT		✗
Verify Step-by-Step [1]	Best-of-N	Per-step process reward annotated by human	✓
MATH-SHEPHERD [12] / pDPO [13]	Best-of-N	Per-step process reward inferred from random rollout	✓
TS-LLM [14]	MCTS	Per-step process reward inferred from TD- λ [15]	✓
V-STaR [7]	CoT	Final outcome reward generated by multi-iteration LLMs	✓
Self-Rewarding [16]	CoT	Final outcome reward generated and judged by LLMs	✗
ReST-MCTS* (Ours)	MCTS*	Per-step process reward inferred from tree search (MCTS*)	Multi-Iter

ReST-MCTS*: LLM Self-Training via Process Reward Guided Tree Search

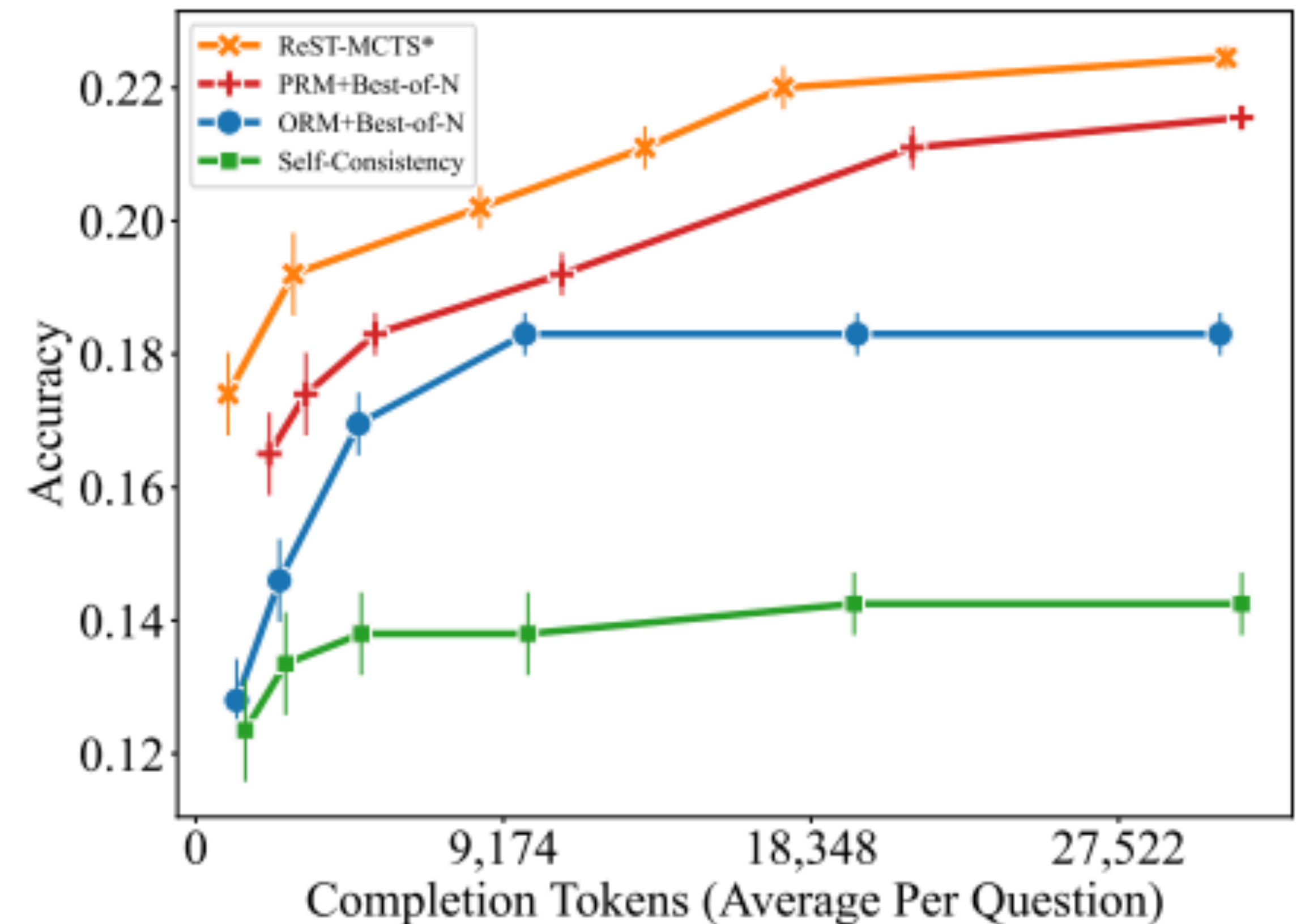
Table 2: Primary results by training both policy and value model for multiple iterations. For each backbone, different self-training approaches are conducted separately. This means each approach has its own generated train data and corresponding reward (value) model. Our evaluation is zero-shot only, the few-shot baseline only serves as a comparison.

Model	Self-Training Methods	MATH	GPQA _{Diamond}	CEval-Hard	Ave.
LLaMA-3-8B-Instruct	0th iteration (zero-shot)	20.76	27.27	26.32	24.78
	0th iteration (few-shot)	30.00	31.31	25.66	28.99
	(Below are fine-tuned from model of previous iteration with self-generated traces)				
	w/ ReST ^{EM} (1st iteration)	30.84	26.77	21.05	17.22
	w/ Self-Rewarding (1st iteration)	30.34	26.26	25.66	27.42
	w/ ReST-MCTS* (1st iteration)	31.42	24.24	26.97	27.55
	w/ ReST ^{EM} (2nd iteration)	33.52	25.25	21.71	26.83
	w/ Self-Rewarding (2nd iteration)	33.89	26.26	23.03	27.73
	w/ ReST-MCTS* (2nd iteration)	34.28	27.78	25.00	29.02

ReST-MCTS*: LLM Self-Training via Process Reward Guided Tree Search



(a) Self-training of value model on MATH.



(b) Comparison of value model on SciBench.

Figure 2: Accuracy of different searches on MATH and SciBench with varied sampling budget.

